

Individual Research Project Report

Rocket Exhaust CFD Modelling in Fluent

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Abstract

The exponential growth of the space industry has raised questions about the environmental impact of rocket launches. However, due to a lack of concern for many years, there is limited research and regulation on the subject. This study uses computational fluid dynamics (CFD) simulations in Ansys Fluent to provide an initial assessment of rocket plume behaviour during liftoff in the troposphere. Two case studies were conducted. The first modelled the exhaust plume of a liquid hydrogen and liquid oxygen rocket to investigate the interaction between combustion products and the surrounding atmosphere at various altitudes. The results reproduced key physical features, such as shock and expansion waves. However, limitations related to boundary conditions and species distribution were also highlighted, particularly at higher altitudes, where reduced air density lessens plume-air interactions. The second case study extended the analysis across multiple altitudes, using parameter sets to vary atmospheric pressure and temperature. However, due to the high computational cost, the outputs were restricted to a single parameter (mass flow rate), which remained constant across altitudes due to the unchanged inlet conditions.

These findings demonstrate the potential and challenges of modelling rocket emissions with CFD. While the simulations captured essential physical mechanisms, further refinements are required to address the effects of boundary conditions, species interactions and computational efficiency. This work lays the groundwork for more advanced studies on the environmental impact of rocket emissions in the upper atmosphere.

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1 Introduction

In the mid-20th century, the space industry began. It was marked by a race into space exploration. From the first artificial satellite, the first man to space, the first steps on the moon, to the first space stations ; the space industry became a key sector for navigation and scientific discovery. During this period, the environmental impact of rocket launches received little attention from the scientific community [10]. It was generally assumed that the number of launches was too small for rocket emissions to significantly affect the atmosphere [22]. However, in recent years, there has been a rapid growth in the space industry. It has been marked by an increase in the frequency of rocket launches due to the attraction to the low Earth orbit (LEO). This has impacted the environment, especially the atmosphere [22, 10, 11, 19, 31, 24]. The emissions released by rockets pass through the different layers of the atmosphere and affect the environment. As launch frequencies continue to rise, understanding the rocket emissions during its ascent and their effects has become crucial.

1.1 Exponential Rise in Space Industry

In recent years, the space industry has experienced a comeback after a period of stagnation in the early 2000s. This rise is mostly driven by the growing demand for satellite-based services in LEO. These services include studying Earth’s climate and environment, supporting communication networks, navigation systems, scientific research, and space resource exploration [31, 24, 12].

The expansion of the market has been through the commercialisation of rockets with the involvement of the private sector. In the past, the space industry was primarily funded by the government [19]. The interest of the private sector has led to technological breakthroughs in rocket and satellite development, thus reducing manufacturing costs [24]. These companies mainly offer rocket launches for satellites. However, a sub-market that will emerge in the future is space tourism, which has already begun with Blue Origin’s trips [11].

As of today, there are around 10,000 satellites in space, of which around 7,000 are still functioning [12], with most of the traffic concentrated in LEO [19]. Estimation of the industry growth varies, but forecasts suggest that there could be between 20,000 to 50,000 satellites in orbit by 2030 and up to 60,000 by 2040, with as many as 10,000 launches occurring annually [22, 19, 20]. These figures highlight the rapid expansion of satellite constellations in LEO.

1.2 Economy of the Space Industry

A rise in the industry means a rise in business opportunities. Therefore, to meet the demand, there is a growth of private companies and startups in the industry. In Europe, companies such as Isar Aerospace in Germany offer launch services for small and medium-sized satellites [26]. Sirius, a French company, is developing affordable and sustainable launch solutions [27]. Another example is Maia Space, which combines space launches with sustainability by offering a reusable, eco-friendly mini launcher [28]. Outside

of Europe, the Japanese company Space One is contributing to the industry’s growth with its launch vehicle named ”Kairos”. These satellites are typically placed in LEO or sun-synchronous orbit, depending on their intended purpose [9].

On a larger scale, major companies like SpaceX have ambitious goals for satellite deployment in LEO. Their target is to launch up to 42,000 satellites, with around 5,000 already in orbit as of early 2025. Other companies, such as OneWeb and Amazon’s Project Kuiper are also planning to deploy several hundred satellites in this orbit. Meanwhile, China is planning to enter the industry by launching its own satellite constellations [31].

The space industry is projected to grow significantly, from \$0.57*trillion* in 2023 to \$1.8*trillion* by 2035 [22]. The United States is expected to hold the largest share of this market [32, 24] ; although new players such as China, India, and Japan are also emerging [31]. This growth is largely driven by reduced launch costs and the commercialisation of space activities [24]. It is important to note that the cost of manufacturing and deploying a satellite can range from \$50*million* to \$400*million* [22].

While the growth of the space industry is beneficial for the economy [23], it is also raising concerns about its environmental impact. Rocket’s emissions and debris generation will have consequences on our planet and its inhabitants. The commercial activity accelerates the process. It is crucial to assess, manage and regulate these impacts for the sustainability of the industry and our good.

1.3 Environmental Impact and Concerns of Rocket Launches

The environmental impact of rocket launches is various and raises serious concerns. Rockets are unique in that they are the only human-origin emissions released directly into all layers of the atmosphere [22, 12, 16]. For a long time, the long-term effects of these emissions were either not considered at all, or not fully understood — particularly the indirect impact.

One of the most critical concerns is about the ozone layer located in the stratosphere. This layer plays a vital role in protecting the Earth from harmful ultraviolet radiation. It is continuously monitored, but this needs to be more closely monitored due to the expected scenario for the space industry. Even small reductions could have significant consequences.

Rocket emissions not only affect the ozone layer but also have broader impacts on the climate and ecosystems. For instance, booster stages are often intentionally dropped into the ocean [12], contributing to marine pollution and habitat disruption. There is also evidence suggesting that rocket activity may contribute to the formation of mesospheric clouds [10, 11], though more research is needed to fully understand it.

Sky brightness is another issue that can affect ecosystems and even humans. The growing number of satellites increases the amount of reflected light in the night sky, raising its brightness by about 10% above natural levels [12, 19]. This disrupts nocturnal wildlife, including animals and insects that rely on celestial cues for navigation and migration. Artificial light pollution could have long-term ecological impacts [19].

Another principal environmental issue is space debris. In recent years, debris has become a growing concern due to increasing congestion in orbit [12, 19]. The problem

is that it contributes to space pollution and uses up orbital space. The accumulation of space debris is expected to increase due to more frequent launches and the Jevons paradox. The paradox is that even without new launches, collisions between existing objects can cause debris to increase [19, 31]. These collisions generate more fragments, creating a vicious cycle [11].

The main danger of debris is its velocity, which poses a threat to operational satellites, potentially damaging or destroying them [31]. Space debris is unsustainable, as it can remain in orbit for hundreds of years and reduce the lifespan of other satellites. While some solutions have been proposed, such as debris removal systems, most target only defunct satellites [19]. Even though some rockets are now partially reusable, parts of them still end up as debris [12].

The final concern regarding debris is its re-entry into the atmosphere. While most debris disintegrates during re-entry, this is a part of rocket emissions that is often overlooked [20]. One study found that approximately 10% of particles from space debris re-entering the atmosphere are present in the natural stratospheric sulphate layer [22].

A final issue that will impact the environment is the development of space tourism, which is set to become a major industry in the coming years [11]. The carbon footprint of a single passenger on a space tourism flight is estimated to be equivalent to thousands of flights between Los Angeles and London [19]. The attractiveness of such things is alarming, given that space tourism is set to become more popular and accessible in the future.

These concerns go beyond environmental issues. They also touch on the sustainability and security of the space industry itself [31]. Due to the limited amount of research on the detailed environmental effects of rocket launches, both direct and indirect consequences may remain overlooked. Without proper study and regulation, the rapid growth of space activity could lead to significant long-term problems.

1.4 Rocket Propellants and their Emissions

The primary source of emissions from rockets is the propulsion systems. The propulsion works by combining a fuel with an oxidiser, which reacts to produce thrust. Most rockets operate in multiple stages, typically with a first-stage engine used during lift-off and a second-stage engine used once the rocket reaches higher altitudes. Each stage may use different types of propellants, and the environmental impact varies depending on the combination and ratio of fuel and oxidiser.

Solid propellants were historically among the most widely used [24]. They were preferred for their simplicity, ease of storage, and reliability. However, they cannot be turned off once ignited [10]. Solid motors typically use a combination of aluminium as the fuel and ammonium perchlorate as the oxidiser [19]. While effective, these propellants release aluminium oxide and chlorine compounds, which can be harmful to the atmosphere, especially the ozone layer [16].

In recent years, there has been a shift towards the use of liquid propellants, which offer greater control and efficiency. The most common are kerosene ($RP - 1$), liquid hydrogen (LH_2), methane (CH_4), and hypergolic fuels [16]. These fuels are generally mixed with liquid oxygen (LO_2).

- Kerosene is commonly used in the first stage of launch vehicles due to its high density and ease of handling. However, it produces soot and carbon dioxide emissions.
- Liquid hydrogen is considered the cleanest propellant in terms of emissions, by producing primarily water vapour [11]. It is widely used in upper stages and is considered a promising long-term solution.
- Methane is gaining interest as a cleaner alternative to kerosene, yet it is not widely used.
- Hypergolic fuels, such as dinitrogen tetroxide (N_2O_4) mixed with unsymmetrical dimethylhydrazine (UDMH), ignite on contact and are often used for manoeuvring systems.

Another category is hybrid engines, which combine a liquid oxidiser (often LOX or nitrous oxide) with a solid hydrocarbon-based fuel. These systems aim to balance the advantages of solid and liquid propellants but are still in development stages for broader applications.

According to recent studies, liquid hydrogen appears to have the lowest environmental impact among the major rocket propellant options [11]. This is primarily because it produces clean combustion products, mainly water vapour, which contribute minimally to atmospheric pollution or ozone depletion.

Overall, it is estimated that rockets consumed approximately 40 kilotons of propellant globally in 2014 [23]. A figure expected to rise significantly with the increasing frequency of launches in the coming years. Looking ahead, it is expected that liquid hydrogen and kerosene will remain the dominant propellants in the near future [10], although the industry is actively exploring greener alternatives.

1.5 Lack of Regulation on Rocket Emissions

Currently, there are no binding international regulations that specifically address rocket emissions. Since its establishment in 1959, the Committee on the Peaceful Uses of Outer Space (COPUOS) has brought together diplomats, scientists and experts to discuss the challenges and risks associated with space exploration. Some historical agreements, such as the *1963 Treaty Banning Nuclear Weapon Tests in the atmosphere, outer space, and underwater* impose limits on activities such as nuclear testing in space. However, its treaties tend to focus more on the peaceful use of outer space than on environmental concerns on our planet [31].

Most measures are limited to non-binding guidelines, frameworks, and recommendations. Topics covered include space debris mitigation, the use of nuclear power sources in space and ensuring the long-term sustainability of activities in outer space. However, they have failed to establish enforceable standards or protocols due to the limited knowledge on the subject.

A 2023 United Nations report highlighted the urgency of addressing the environmental impact of increasing launch frequency. The report proposed two possible paths forward [31]:

- Establishing a unified regulatory regime focused on space sustainability ;

- Creating governance frameworks that target specific aspects of space activity, such as debris, emissions, and satellite congestion.

Despite growing concerns, there are still no formal regulations that directly address rocket emissions or their full environmental impact. Without coordinated international action, the environmental and sustainability risks associated with space operations may continue to grow unchecked within the current minimal policy framework.

1.6 Purpose of the Study

As outlined, it is essential to understand the impact that rockets can have on the environment before the problem becomes too severe to recover from. Rocket emissions are a crucial but highly complex topic. This study aims to investigate how chemical species interact in the exhaust plume of a rocket during ascent.

The main objectives of this work are to:

- Review the current state of research in the field;
- Gain a clearer understanding of the subject and its underlying complexity;
- Develop a CFD simulation to model the species present in a rocket exhaust plume;
- Extend the simulation across multiple points along the ascent trajectory.

To achieve these goals, two cases are highlighted. The first focuses on a CFD simulation of the rocket exhaust plume at a fixed altitude. For this case, I selected a liquid propellant combination of liquid hydrogen and liquid oxygen, chosen for its relatively simple and well-documented chemical reaction. This scenario provides an initial baseline understanding of plume behaviour. The second case expands the analysis to different altitudes, restricted to the troposphere. In both cases, several simplifications were applied in order to manage the complexity and focus on the most relevant aspects.

The study is structured as follows. A literature review first assesses the current state of research on rocket emissions. Next, a theoretical background outlines the physical and chemical principles underlying the model and the interactions between chemical species. The methodology is then presented, detailing how the simulations were set up. A dedicated section also reflects on the multiple trials conducted, highlighting the challenges encountered and the iterative process required to achieve valid results. Finally, the outcomes of the simulations are examined both quantitatively and qualitatively, offering a comprehensive perspective on rocket exhaust emissions. The study concludes with a summary of findings and recommendations for future work to expand and improve upon this research.

2 Literature Review

2.1 Previous Studies

Review of Computational Studies on Rocket Performance and Environmental Impact

Previous studies using computational software to analyse rockets have initially focused on optimising their aerodynamic performance during the preliminary stages of launcher design. For example, Viviani and Pezzella (2018) conducted a computational fluid dynamics (CFD) simulation to analyse the aerodynamic flow field during the ascent phase of an expendable launch vehicle. Their simulations, run at various Mach numbers, incorporated rocket plume effects with detailed boundary conditions for both the launcher and booster stages. The objective was to improve aerodynamic database development for future designs, and the results provided valuable flow field data relevant to preliminary aerodynamic modelling. [3]

Similarly, Viviani et al. (2016) sought to compile a comprehensive aerodynamic database using steady CFD methods to analyse the aerodynamics of general launchers. They also carried out unsteady CFD simulations to study booster separation dynamics during staging. Their findings supplied essential aerodynamic data and insights into general flight behaviour and critical staging phases. [2]

Current research trends also demonstrate an increasing interest in satellites located in the very low Earth orbit (VLEO). Studies such as those by Jiang et al. (2023) and Walsh et al. (2021) examine the aerodynamic behaviour of satellites in VLEO in order to optimise their structural design and performance, with a focus on managing atmospheric drag at these high altitudes. Both studies employed the Direct Simulation Monte Carlo method (DSMC), though with different approaches. The study by Jiang et al. used DSMC to calculate aerodynamic drag for basic shapes and a conceptual satellite model, comparing the results with free molecular flow theory [13]. Study by Walsh et al. developed a radial basis function (RBF)-based surrogate model trained on DSMC data to optimise aeroshell shapes, offering drag reduction scenarios and inventories for early-stage design. [34]

While most research has focused on improving the performance of rockets and satellites through design, some studies are starting to address the connection between technical performance and environmental sustainability. For example, Tormena et al. (2022) investigated optimisation scenarios for reusable launch vehicles, considering both engineering and environmental factors. The influence of variables such as propellant type, oxidiser–fuel mixture ratio and flight guidance profiles on launch performance and radiative forcing was evaluated. The results indicated that certain parameters, especially the propellant mixture ratio, can significantly influence a rocket’s environmental impact. [30]

Analytical Approaches to Estimating Rocket Emissions

To gain an initial understanding of rocket emissions, several studies have applied analytical methods to estimate their atmospheric impact. For example, Ross and Sheaffer (2014) evaluate the instantaneous radiative forcing (RF) of rocket engine emissions by linearly scaling RF values from various launch vehicles according to their actual propellant loads. The results indicate that rocket exhaust contributes to positive atmospheric radiative forcing, with black carbon and aluminium particles having the most significant

effects. In contrast, carbon dioxide emissions from rockets exhibit a comparatively low radiative forcing effect relative to particle emissions. [23]

Study by Boigt et al. (2013) further examines the impact of rocket emissions by compiling aircraft measurements of rocket exhaust plumes and comparing these local observations with global model simulations. The findings reveal that rocket emissions can have a significant regional atmospheric impact, particularly in the stratosphere, when compared to other natural and anthropogenic sources. [7]

Another important, yet often overlooked, aspect of this issue is the emissions produced during spacecraft reentry. Murphy et al. (2023) investigate the impact of metal vapours released into the stratosphere during reentry. The study identifies over 20 elements originating from spacecraft reentries by analysing the metal content of stratospheric sulfuric acid particles. The results show that the metal mass from reentries exceeds that of cosmic dust for these elements. Furthermore, projections based on expected satellite activity suggest that up to 50% of stratospheric particles could contain reentry, highlighting a potentially significant and growing source of atmospheric pollution. [20]

Adding to this research, Tyler and Bannister et al. (2024) developed a three-dimensional inventory of stratospheric rocket emissions using 2019 launch data. The study provides a comprehensive breakdown of chemically and radiatively active species produced by common rocket propellants and employs global atmospheric models to assess their effects on the ozone layer. [6]

In addition to these analytical approaches, one notable study, Dominguez et al. (2024), performed a Life Cycle Assessment of different reusable launch vehicles in the European space market. The aim was to identify key design factors for reducing environmental impact, considering the entire life cycle (from raw material extraction to end of life). The assessment covered environmental indicators such as climate impact, water depletion, and land use for two major propellant types: liquid hydrogen and liquid methane. A particular strength of this study was its explicit focus on high-altitude emissions and previously overlooked pollutants, which had been insufficiently addressed in earlier research. The results showed that liquid hydrogen has a lower overall environmental footprint than liquid methane, primarily because it does not emit black carbon. However, the study also showed that overlooking certain pollutants in previous assessments resulted in the climate impact being underestimated by two to three times. [11]

Taken together, these studies highlight the complexity and limitations of our current understanding of rocket emissions. While analytical models provide valuable first-order estimates of radiative forcing and pollutant distribution, life cycle assessment approaches offer a broader view of the environmental impact. However, there are still gaps in our knowledge, particularly concerning re-entry emissions, high-altitude chemical processes and the cumulative impact of multiple rockets.

Atmospheric Dispersion and Earth System Modelling Approaches

One of the most common types of simulation used for studying rocket emissions is atmospheric dispersion modelling. This method is used to study the global impact on the climate and how species propagate. This is often performed using the GEOS-Chem chemical transport model. GEOS-Chem is a three-dimensional atmospheric chemistry model based on meteorological data supplied by NASA's Global Modelling and Assimilation

Office (GMAO).

In a study by Ryan et al. (2022), the model was used to assess the atmospheric and climate impacts of modern rocket launches and spacecraft reentries, with a focus on stratospheric ozone depletion and radiative forcing. The researchers developed an inventory of 2019 launches and reentry debris, as well as a speculative space tourism scenario, and implemented these into GEOS-Chem coupled with a radiative transfer model. Results indicated that rocket emissions could undermine the success of the Montreal Protocol, with ozone loss from chlorine-based compounds and reentry-related emissions particularly affecting the upper stratospheric Arctic. Furthermore, scenarios of space tourism emissions were found to be substantially greater than soot contributions from both surface and aircraft sources. [24]

While GEOS-Chem has been widely applied to rockets, it is also used for other atmospheric contexts. For example, June et al. (2025) used the model, coupled with TOMAS aerosol microphysics, to study the impact of biomass burning plume injection height on aerosols, air quality, and climate. They tested three scenarios with varying injection heights and found that higher plume altitudes increased aerosol concentrations, had minimal effects on surface particulate matter, and enhanced the cooling effect of biomass burning through both direct and indirect radiative mechanisms. [14]

Another modelling software used to model rocket emissions is the NASA GISS ModelE Earth System Model. Tsigaridis et al. (2024) used this model to simulate and evaluate the atmospheric impact of launch vehicle emissions in future methane-fueled rocket scenarios. They compared two launch frequency scenarios: 1,000 launches per year and 10,000 launches per year. The simulations analysed gaseous emissions: CO , NO_x , H_2O , as well as black carbon. The results indicated that black carbon emissions had the most significant atmospheric impact, while gaseous emissions were comparatively negligible. Although methane-fuelled rockets were found to be more efficient overall, their aerosol emissions remain a significant environmental concern. [17]

In summary, these modelling approaches provide insights into pollutant transport, chemical interactions and climatic effects. They combine first-order analytical estimates with a detailed and global atmospheric model. They reveal that, regardless of fuel type, soot and particle matter are the dominant cause of the atmospheric impact of rocket operations, particularly in scenarios involving a high launch frequency, such as those projected for space tourism.

Rocket Exhaust Modelling and Environmental Impacts

Another important research focus lies in modelling rocket exhaust and its interactions with the atmosphere. While approaches vary, these studies collectively reveal the complexity of exhaust behaviour, from chemical reactions in the nozzle to large-scale atmospheric effects.

For example, study by Kokkinakis and Dimitris (2022) used high-order CFD methods and implicit large-eddy simulations to model a Falcon 9-like vehicle, tracking exhaust gas dispersion and heat transfer up to 67 km altitude. The results showed significant CO_2 injection into the mesosphere, equivalent to $26km^3$ of atmospheric air, and persistent thermal NO_x formation even at high altitudes, highlighting the climate impact of frequent launches[15]. In contrast, Chen et al. (2011) modelled the combustion of a hybrid

propulsion system to improve the understanding of the internal flow and reaction processes. By comparing with experimental data, predictions can be made for the future [8].

While these two studies examined near-field exhaust behaviour, study by Bowen et al. (2020) widened the scope to include the upper atmosphere. The Global Ionosphere Thermosphere Model was modified to simulate the depletion of ionospheric plasma by rocket exhaust species. When applied to the Jason-3 and FORMOSAT-5 launches, the model successfully reproduced the observed patterns of ionospheric holes, although it overestimated their magnitude. This indicates both progress and remaining uncertainties in modelling the interaction of large-scale plumes [5].

Schuch and Fisch (2018) focused on health impacts at the surface, adapting the Weather Research and Forecasting model to assess the dispersion of toxic gases (notably HCl and CO) from a Brazilian satellite launcher. The simulations revealed that critical exposure levels could be reached within the first ten minutes after launch, and that wind conditions would determine whether nearby populations faced serious risks. [25]

Other research targeted propulsion-specific optimisation rather than environmental outcomes. Study by Vestnes (2016) used ANSYS Fluent to simulate the complex multiphysics processes occurring within a hydrogen peroxide thruster. This involved combining Eulerian multiphase modelling, porous media pressure drop calculations, heterogeneous reaction kinetics and turbulence modelling. This demonstrated the software’s potential to replace some costly experimental testing [33]. Similarly, Naga et al. (2021) investigated combustion instability in gas turbine combustors. They integrated control theory with CFD simulations to evaluate performance and emissions across different fuels and flow conditions. These methodologies could potentially be transferred to the analysis of rocket engine stability [21].

Together, these studies demonstrate various applications of modelling, ranging from the behaviour of micro-scale catalysts to the effects on the global atmosphere. They show how exhaust modelling can support technical optimisation and reveal environmental and health impacts at vastly different spatial and time scales.

2.2 Knowledge Gaps in Rocket Emissions Research

There is still a significant lack of knowledge regarding rocket emissions and their environmental impact [12, 19]. This issue was underestimated for a long time and received limited attention from researchers and policymakers. Even now, despite growing awareness, the study of rocket emissions remains complex and underdeveloped.

One major challenge is the lack of scientific research. Few studies have been conducted to fully understand the consequences of rocket launches on the atmosphere and climate. One reason for this is that concerns about climate change are a relatively recent phenomenon [19]. This under-representation makes it difficult to form detailed conclusions or develop environmental policies.

A second challenge lies in the complexity of modelling and simulation [19]. Many complex physical phenomena occur, such as the chemical composition of rocket exhaust, reactions at high altitudes and interactions with various layers of the atmosphere. These are extremely difficult to simulate accurately. These processes involve high-energy reactions,

rare chemical species, and dynamic conditions that are hard to reproduce in laboratory settings.

Moreover, the scarcity of data further complicates the situation. There is limited experimental data available for both atmospheric rocket emissions and the effects of space debris re-entering the atmosphere, which can also release particles into the upper atmosphere [24]. The distribution of these emissions throughout the different layers of the atmosphere is not well understood, and current models often lack the volumetric distribution data necessary for accurate evaluations.

Life Cycle Assessments, which evaluate the total environmental impact of a rocket over its entire lifespan, from production to disposal, are also rare [19]. Without such assessments, it is difficult to quantify the long-term sustainability of different propulsion systems and launch practices.

Regarding the propulsion system, research into the impact of liquid propellants is still in its early stages, as this is a new approach for generating thrust in rockets. Solid propellant was the standard option, but it generated too many harmful emissions. In this new approach, liquid propellants are slowly replacing them, but environmental research needs to be conducted to ensure there is no direct or indirect impact.

In summary, the field suffers from a lack of studies and limited data, as well as high modelling complexity. These factors prevent us from understanding the full environmental impact of space activities. Closing these gaps will require targeted research, increased funding and international collaboration. Concerning these knowledge gaps, this study will focus on modelling the exhaust plume of a liquid propellant rocket engine in the troposphere.

3 Methods

3.1 Simplifying the Scope

To achieve the objectives of this study, it was necessary to make several decisions and simplifications to balance physical accuracy with computational viability. This section outlines the main choices made and explains the reasoning behind them. As highlighted in the literature review, different types of simulations can be performed when studying rocket emissions, ranging from detailed exhaust plume modelling to large-scale atmospheric dispersion. However, there is no single open-source software that is currently capable of simulating the entire chain of processes (from exhaust generation at the nozzle exit, through atmospheric propagation, to the global-scale effects of emissions and even re-entry phenomena). Therefore, most studies either combine multiple tools or limit their focus to a specific stage of the problem.

This study focuses purely on the rocket’s ascent through the troposphere. The environmental impact of such a mission is determined by three main categories of parameters.

The first one is the launch vehicle characteristics. These include the rocket’s geometry, weight, payload, and, most importantly, its propulsion system. The type of propellant, oxidiser-to-fuel ratio, propellant flow rate, burn time, and nozzle geometry all strongly influence exhaust composition. Modern rockets can involve multiple booster stages, sometimes using different propellants (e.g., solid propellant for the first stage and liquid propellant for the second, as in Space One’s launcher [9]). Within each stage, several engines may operate simultaneously, and the combined emissions must be accounted for. The second one is flight and operational parameters. These describe the mission profile, such as the trajectory, total flight time, and thrust profile of each booster stage. Emissions are not limited to ascent, as discarded rocket stages re-enter the atmosphere and contribute to environmental impacts.

The third one is atmospheric interaction. As the rocket travels upward, it passes through multiple atmospheric layers, each with distinct densities, temperatures, and chemical compositions. These variations govern how exhaust gases interact with the surrounding environment, either directly or indirectly. A particular concern is the stratospheric layer, where ozone depletion remains a priority issue. However, focusing exclusively on stratospheric effects risks overlooking impacts in other layers.

In practice, each of these parameters involves complex physical processes described by multiple equations and interdependent laws. Capturing all of them simultaneously is expensive computationally. The challenge lies in finding a balance between reducing environmental impact and maintaining simulation reliability.

For this study, the problem was simplified in several ways:

- The analysis is limited to the troposphere, excluding re-entry debris and life-cycle assessment.
- The rocket geometry was reduced to a single nozzle, eliminating the complexity of multiple stages.
- The simulation begins after combustion, at the throat, to avoid modelling in-chamber

combustion chemistry and reduce the number of governing physical laws.

- The ascent was modelled at a set of discrete altitude points rather than a continuous trajectory, which would require significant additional computation and programming.
- The trajectory itself was simplified to a direct vertical ascent rather than a curved profile.
- The number of chemical species included in the exhaust was reduced to maintain reasonable computational costs.

These simplifications were implemented with the sole purpose of reducing computational demand while still capturing the essential physics of the rocket exhaust plume and its interaction with the atmosphere.

Despite these simplifications, the approach favoured in this study still provides valuable insights into the fundamental interactions between rocket exhaust and the atmosphere. By narrowing the scope to the troposphere and focusing on post-combustion exhaust dynamics, the study finds a balance between physical accuracy and computational accuracy. This framework serves as a foundation for progressively more complex simulations, offering a clear path for future work to expand on altitude continuity, real trajectory modelling, and a more complete set of species.

3.2 Theoretical background

Having narrowed down the scope of the study, the next step is to explore the theoretical background that supports the work. This section provides the necessary foundations in propulsion, chemistry and physics to improve understanding of the modelling approach. Beginning with an overview of rocket propulsion systems and the formation of the exhaust plume, then examines the chemical processes that occur in the nozzle through the reaction between fuel and oxidiser. Finally, it introduces the physical principles that govern CFD simulations and illustrates how these principles are applied in this study.

3.2.1 Fundamentals of Rocket Propulsion and Exhaust Flow

The main source of rocket emissions is the propulsion system, which occupies a large part of the vehicle. This system consists of six main components: two tanks for storing the fuel and oxidiser; pumps for drawing the propellants from the tanks; a combustion chamber for igniting the mixture; a throat for accelerating the gases by converting pressure into velocity; and finally, a nozzle for ejecting the exhaust and producing thrust(Figure 1).

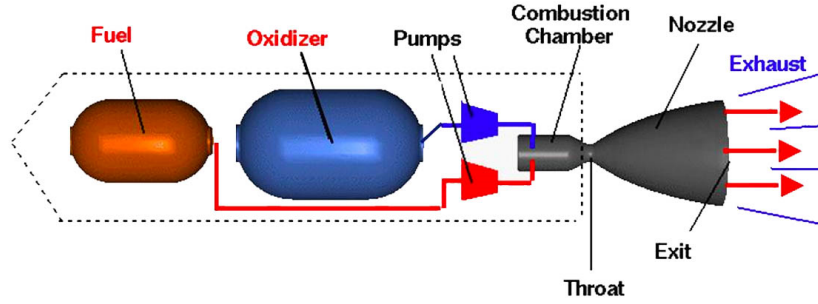


Figure 1: Scheme of the propulsion system of a rocket [1]

This study focuses on the final stages of the process after combustion has occurred by modelling the exhaust gases directly. Figure 2 illustrates the principal physical phenomena at this stage. Following combustion, high-pressure, relatively low-velocity gases are produced. As they pass through the throat of the nozzle, they are transformed into lower-pressure, high-velocity gases, which are then accelerated and directed further by the nozzle. This conversion process is essential for generating sufficient thrust to lift the rocket off the ground.

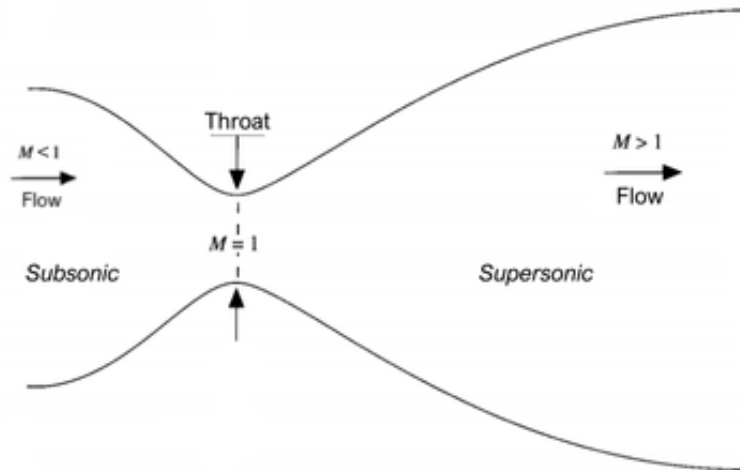


Figure 2: Flow through a nozzle of a rocket engine

3.2.2 Stoichiometric Combustion Parameters

After explaining the propulsion system, it's now the turn to the chemical aspect of how the propellant species interact within the combustion chamber. The chosen propellant is a liquid propellant composed of liquid hydrogen as the fuel and liquid oxygen as the oxidiser. When these reactants are mixed and ignited, a combustion reaction is generated that releases energy and generates high-pressure gases.

The unbalanced reaction can be written as:



Balancing the equation for stoichiometric combustion at high temperature:



In this process, hydrogen and oxygen combine in a precise molar proportion. The stoichiometric molar ratio of oxidiser to fuel is:

$$\frac{\text{oxidiser}}{\text{fuel}} = \frac{n_{O_2}}{n_{H_2}} = 1/2 = 0.5 \quad (3)$$

Using the molar masses of the species:

- $M_{H_2} = 2.m_H = 2 * 1.01 = 2.02g/mol$
- $M_{O_2} = 2.n_O = 2 * 16 = 32g/mol$

The mass ratio between oxidiser and fuel can be obtained:

$$r_m = \frac{m_{O_2}}{m_{H_2}} = \frac{n_{O_2}.M_{O_2}}{n_{H_2}.M_{H_2}} = \frac{1 * 32}{2 * 2.02} = 7.92 \quad (4)$$

This means that, under stoichiometric conditions, approximately 7.9 kg of oxygen are required for every 1 kg of hydrogen. These ratios are fundamental when setting boundary conditions in the CFD model, as they define the relative mass flow rates of fuel and oxidiser. They give a strong influence on the combustion temperature, exhaust gas composition, and plume expansion dynamics, which together determine the efficiency and environmental footprint of the rocket.

3.2.3 Foundations for Rocket Exhaust Simulation

Having examined the underlying composition and chemistry of the propulsion system, it is important to understand the physical theoretical framework that governs the behaviour of exhaust gases once they leave the nozzle. This is where CFD comes into play, providing the tools to simulate the exhaust plume during the rocket's ascent. The theory behind CFD explains what is being modelled, as well as the physics that control the motion, mixing and chemical transformation of the gases.

At the core of CFD are the Navier–Stokes equations, which describe how fluid properties evolve in space and time under the influence of various forces, such as gravity, pressure, and viscosity. These can be divided into three fundamental conservation laws:

1. Continuity (mass conservation)

$$\frac{d}{dt} \int_V \rho dV + \int_{\partial V} \rho \mathbf{u} \cdot \mathbf{n} dA = 0 \quad (5)$$

This ensures that mass is neither created nor destroyed within the control volume. Here, ρ is the fluid density, $\mathbf{u}$ is the velocity vector, V is the control volume, and $\mathbf{n}$ is the outward-pointing unit normal vector on the surface ∂V .

2. Momentum conservation (Newton's second law)

$$\frac{d}{dt} \int_V \rho \mathbf{u} dV + \int_{\partial V} \rho \mathbf{u} (\mathbf{u} \cdot \mathbf{n}) dA = \sum F \quad (6)$$

This relates changes in momentum to the sum of forces F acting on the fluid, including pressure, viscous, and body forces.

3. Energy conservation (first law of thermodynamics)

$$\frac{d}{dt} \int_V \rho E dV + \int_{\partial V} \rho E \mathbf{u} \cdot \mathbf{n} dA = \dot{Q} - \dot{W} \quad (7)$$

Here, E represents the total energy per unit mass, $\dot{Q}$ is the heat transfer rate, and $\dot{W}$ is the work done by the system.

In addition, the species transport equation is crucial because it accounts for the chemical reactions between exhaust gases:

$$\frac{d}{dt} \int_V \rho Y_i dV + \int_{\partial V} \rho Y_i \mathbf{u} \cdot \mathbf{n} dA = \dot{w}_i \quad (8)$$

where Y_i is the mass fraction of species i and $\dot{w}_i$ is its production or consumption rate due to chemical reactions.

Together, these equations form the mathematical foundation for modelling rocket exhaust plumes. They are directly connected to the design of the propulsion system, since the geometry of the nozzle, the combustion conditions and the fuel chemistry all define the initial conditions that are entered into the CFD simulation. Ultimately, they determine how the emissions disperse and impact the atmosphere.

3.2.4 Atmospheric Parameters During Rocket Ascent

The final theoretical framework used in this study concerns the atmospheric conditions encountered during ascent. Since the focus is limited to the troposphere, only a few key parameters were considered to simplify the problem while still capturing the dominant effects.

The first varying parameter is the atmospheric pressure, which decreases exponentially with altitude according to the barometric equation:

$$p = p_0 \exp \left(-\frac{Mg}{RT_0} z \right) \quad (9)$$

where p_0 is the atmospheric pressure at sea level ($101325Pa$), M is the molar mass of air ($0.029kg/mol$), g is the gravitational acceleration ($9.81m/s^2$), R is the universal gas constant ($8.314J/(molK)$), T_0 is the surface temperature, and z is the altitude in meters. Since pressure decreases with altitude, the nozzle exit pressure ratio also changes continuously during ascent.

The second varying parameter is the temperature, which follows a linear lapse rate relationship in the troposphere:

$$T = T_0 - L * z \quad (10)$$

where L is the temperature lapse rate (typically $6.5K/km$ in the standard atmosphere).

Finally, for simplicity, the atmospheric composition was assumed constant across all altitudes in the troposphere, with fixed mass fractions of nitrogen ($N_2 = 0.79$) and oxygen ($O_2 = 0.21$).

Now that these atmospheric variations have been defined, the study can transition from the theoretical background to the CFD setup and simulation methodology.

3.3 Simulation setup

The ascent of a rocket in the troposphere was modelled using ANSYS Workbench and Fluent to assess emissions along the trajectory. The selected propellant combination was liquid hydrogen and liquid oxygen, due to the simple reaction and frequent use in modern launch vehicles.

ANSYS is a popular and robust platform for engineering simulations that allows geometry, meshing and CFD to be integrated into a single environment. The CFD solver Fluent enables the modelling of multiphysics interactions, such as combustion products, heat transfer and plume expansion. This numerical approach offers a cost-effective and practical alternative to full-scale experimental testing while providing insight into the complex physical and chemical processes that govern rocket exhaust emissions.

3.3.1 Geometry Definition

The first step in the simulation was the creation of the geometry. I modelled half of a rocket nozzle from the throat, taking advantage of its symmetry to reduce computational cost. The geometry was kept simple, with dimensions inspired by values found in the literature (Tab. 1). The main constraints applied are presented below, with dimensions given in millimetres (Figure 3).

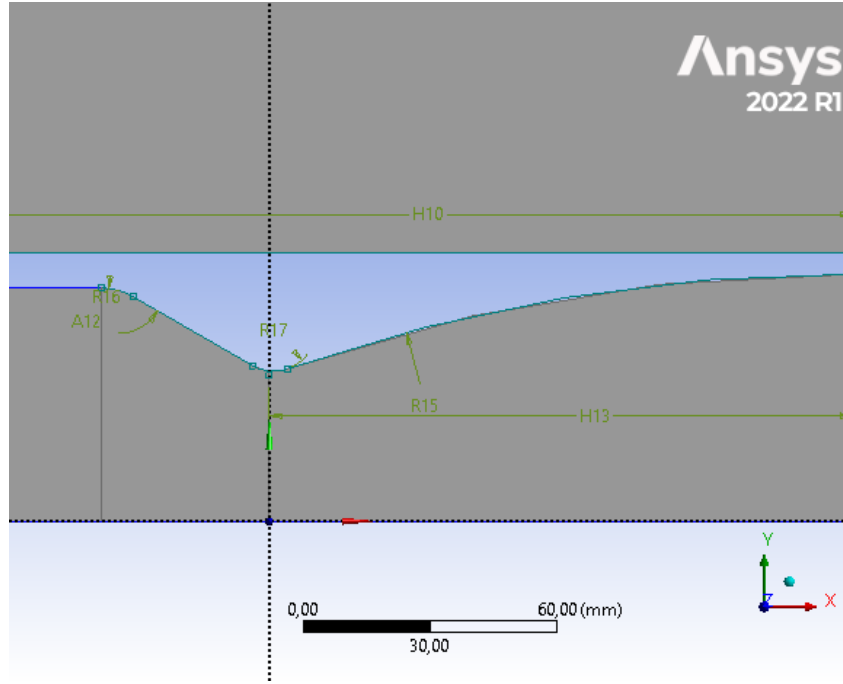


Figure 3: The geometry of the nozzle

Variable	Dimension
A21	150°
H10	210 mm
H13	137.55 mm
R15	400 mm
R16	15 mm
R17	10 mm
V14	5 mm
V21	55 mm

Table 1: Geometric dimensions of the nozzle

Next, I defined the computational domain surrounding the nozzle (Figure 4). The domain was designed to be large enough to allow proper visualisation of the rocket exhaust plume, but not excessively large so that fine mesh refinement could be applied without incurring significant additional computational costs.

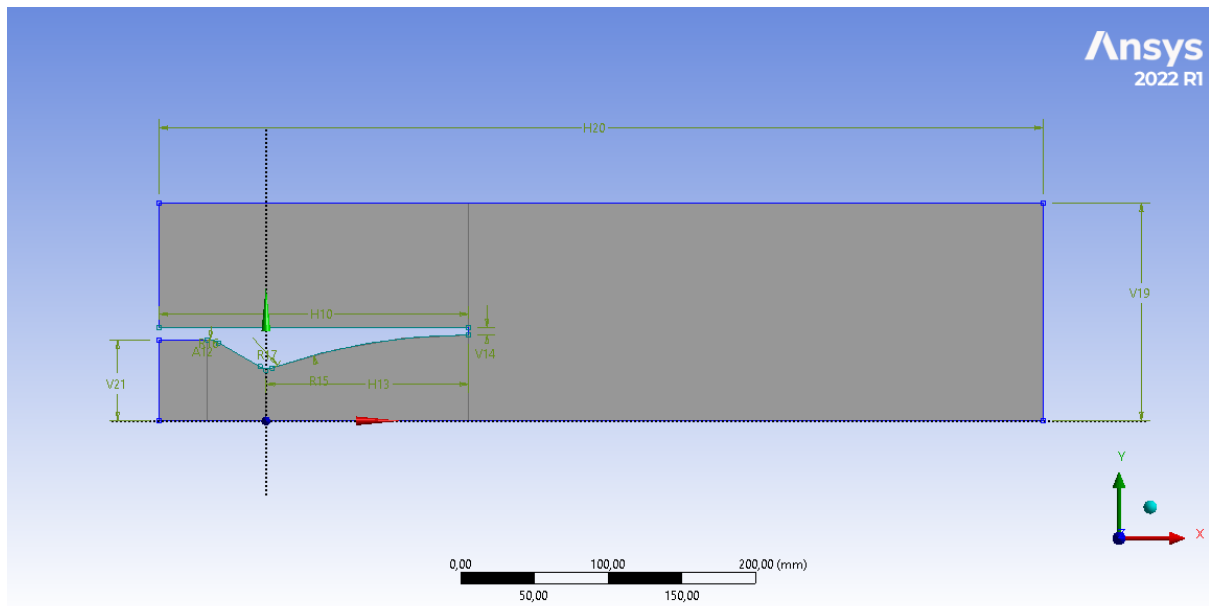


Figure 4: Computational domain geometry

Variable	Dimensions (mm)
H20	600
V19	147.87

Table 2: Dimension of the computational domain

Finally, I named the main parts of the geometry to define the boundary conditions (cf. Figure 5). The farfield represents the surrounding atmosphere, the inlet corresponds to the exhaust gas flow after combustion, and the wall defines the nozzle surface.

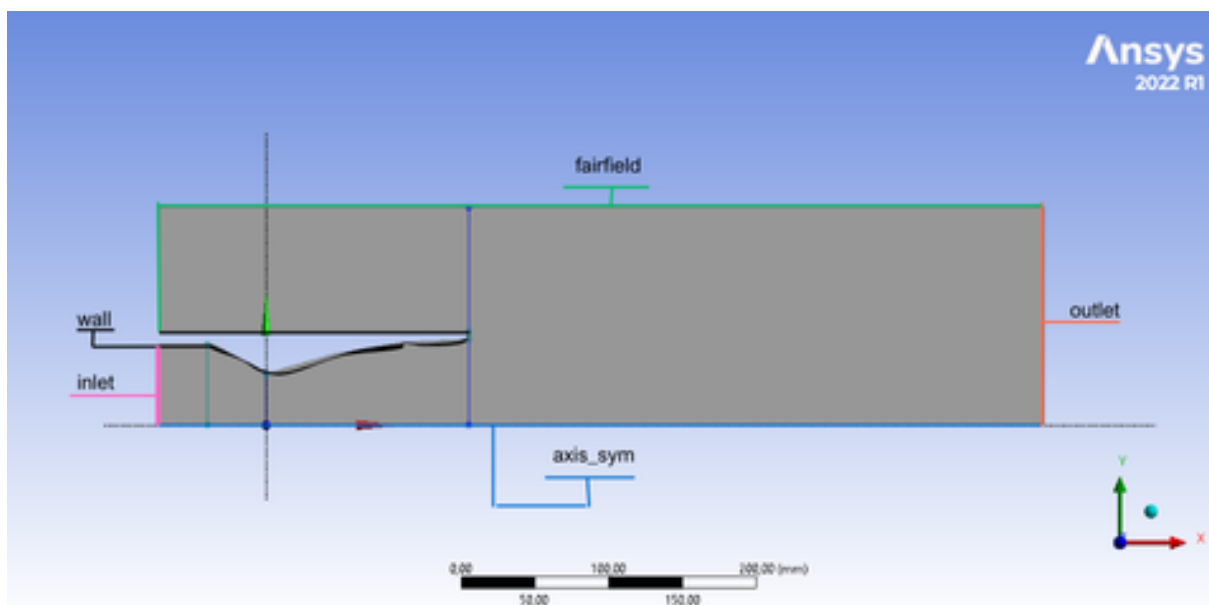


Figure 5: Geometry part name

3.3.2 Mesh Generation

After creating the geometry, the next step was the meshing process. Meshing is critical because it determines how much detail of the flow field can be captured while balancing the computational cost. A finer mesh gives more accurate results but at the price of longer computation times, while a coarse mesh reduces cost but may lose essential physical details. However, if the mesh is too coarse, it could lead to a divergence in the results. It is all about a question of balance between quality and quantity.

To obtain a proper mesh, I first created an additional sketch in the geometry and introduced splitting lines. These lines divided the surface into smaller regions, which helped generate a more structured and uniform mesh (Figure 6).

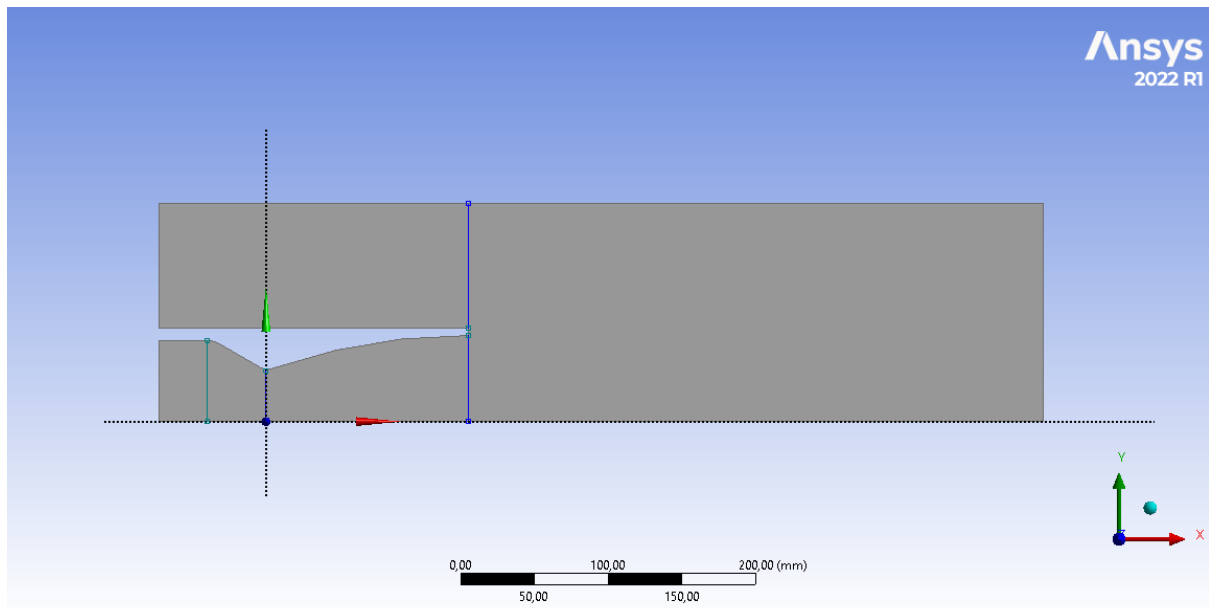


Figure 6: Sketch to split the surface faces

Then, I refined the mesh using several tools such as face meshing and sizing controls, which allowed me to concentrate resolution around critical areas, like at the exit of the nozzle. The final mesh contained more than 60000 elements (Figure 7).

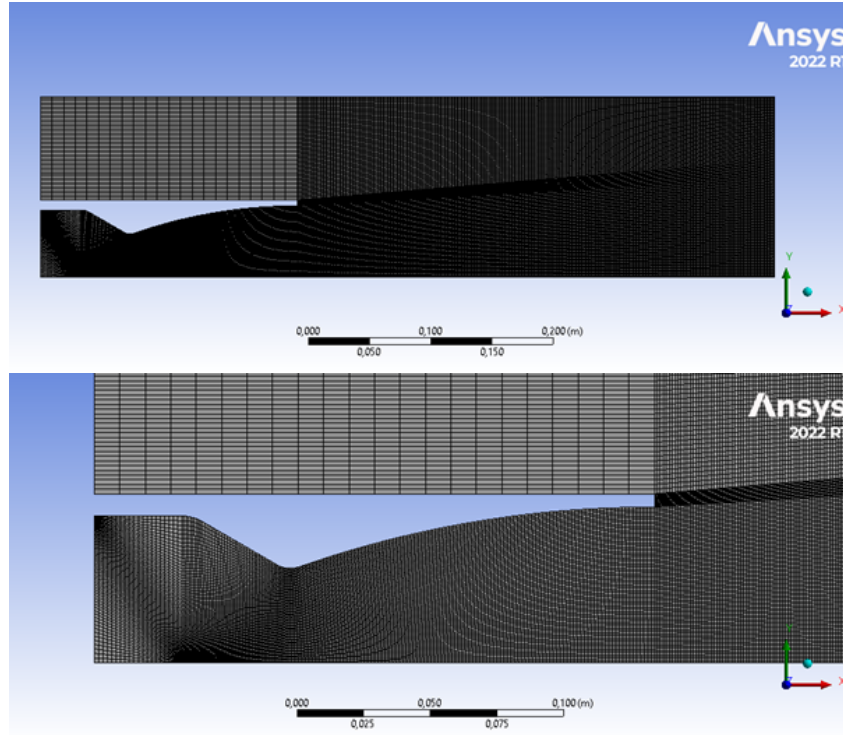


Figure 7: Mesh of the nozzle

To evaluate mesh quality, I inspected two key metrics: orthogonal quality and aspect ratio. Orthogonal quality measures how close the angle between adjacent elements is to the ideal, with values ranging from 0 (poor) to 1 (perfect). Aspect ratio evaluates how stretched an element is compared to an ideal shape; values close to 1 are best.

In practice, acceptable guidelines are:

- Orthogonal quality > 0.2
- Aspect ratio < 5

In our case, the mesh is quite good. This provides a good compromise between capturing relevant physics and maintaining reasonable computational costs.

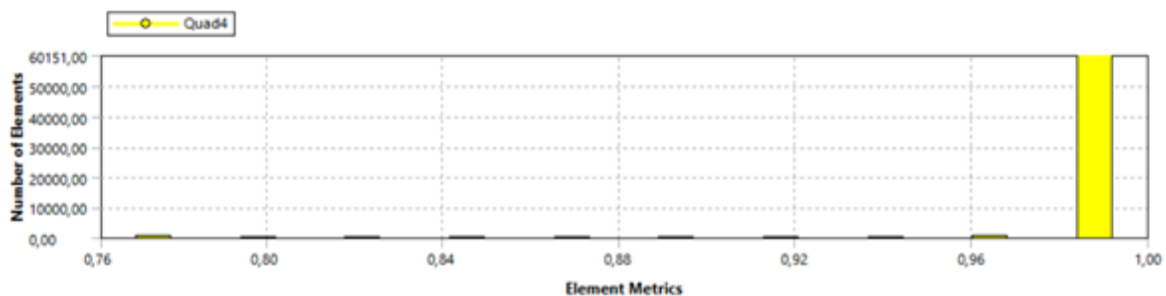


Figure 8: Orthogonality quality of the mesh

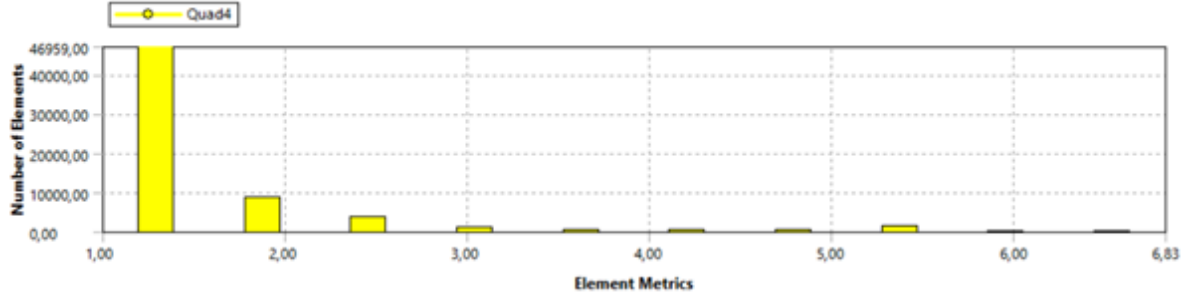


Figure 9: Aspect ratio of the mesh

3.3.3 Modelling Approach

After completing the mesh, the next step was to select the modelling strategy for the simulation. Two possible approaches were considered:

- Before combustion, where the liquid fuel and oxidiser would be injected directly from their tanks into the combustion chamber, leading to the chemical reaction. This strategy requires capturing the complex multiphase and reactive flow physics involved in combustion.
- After combustion, where the chemical reaction is assumed to have already taken place inside the combustion chamber. The simulation then starts with combustion products serving as the inlet condition. This reduces complexity by focusing on the exhaust expansion through the nozzle and into the atmosphere.

In the pre-combustion case, the simulation would require a pressure-based solver, accounting for multiphase flow and species transport to capture the mixing and ignition of liquid fuel and oxidiser. A turbulence closure such as the $k-\omega$ model would also be necessary to resolve the interaction between phases.

In the post-combustion case, the assumption is that the chemical reaction has already occurred. The simulation can therefore be simplified to a density-based solver, without the need for multiphase or pre-reaction species transport.

For this study, the post-combustion approach was chosen. This simplification avoids the additional computational cost and convergence difficulties of simulating the combustion process, while still allowing analysis of the exhaust plume and its environmental implications. Since the flow is supersonic, a density-based solver was employed with a 2D axisymmetric planar configuration. To account for viscous effects, a $k-\omega$ SST (Shear Stress Transport) turbulence model was used, alongside the energy equation and species transport for a mixture model, ensuring that thermal and compositional effects of the exhaust gases were included.

3.3.4 Species Definition

In order to model the interaction between the rocket exhaust plume and the surrounding atmosphere, chemical species from both sources had to be included: those formed by combustion and those that are naturally present in the environment. Hence, I used NASA's Chemical Equilibrium with Applications (CEA) program. This tool calculates

chemical equilibrium for combustion processes and provides both physical properties (such as pressure and temperature in the combustion chamber and throat) and chemical properties (such as the mass fractions of individual species).

The CEA results showed that the main product of the hydrogen–oxygen combustion is water vapour, which is consistent with the theoretical chemical reaction. However, the program also indicated the presence of additional minor species, such as hydroxyl (OH), hydrogen (H_2), oxygen (O_2) and atomic oxygen (O). To account for these findings, I expanded the model beyond water vapour alone to include these extra products.

The final mixture, referred to as exhaust, therefore consisted of:

- Oxygen molecule (O_2)
- Water vapour (H_2O)
- Nitrogen (N_2)
- Hydroxyl (OH)
- Hydrogen (H_2)
- Atomic oxygen (O)

Therefore, the mixture captures the primary combustion and reactive intermediates at the same time. This is highly relevant when modelling and understanding plume-atmosphere interactions.

To represent the thermophysical properties of the exhaust, I modelled the density as an ideal gas and the viscosity using Sutherland’s law. This approach is more physically accurate for hot exhaust gases, since it accounts for viscosity variation with temperature.

For Sutherland’s law, I used the three-coefficient formulation with the following values:

- Reference viscosity = $1.716e - 5 Pa$
- Reference temperature = $273.15 K$
- Sutherland constant = $110.4 K$

3.3.5 Boundary Conditions Setup

The next step was to set the boundary conditions. In preparation for this, I had already named the different parts of the geometry (Figure 5).

For the inlet, I assumed that the pressure inside the throat was 5 MPa. To determine the corresponding thermodynamic properties and species composition, I used CEA program (Appendix 1).

The outlet was defined as a pressure outlet. In first case study, the temperature was set to 281.5 K at 1 km of altitude and 223 K and at 10 km in altitude for the second scenario. Since the simulation domain was restricted to the troposphere, the atmospheric composition was simplified to include only two species: nitrogen (79%) and oxygen (21%) by mass fraction.

Parameter	Value
Gauge Total Pressure	2 897 400 Pa
Supersonic/Initial Gauge Pressure	2 887 400 Pa
Temperature	3380 K
Mass fraction of H_2O	0.816
Mass fraction of O_2	0.067
Mass fraction of N_2	0.000
Mass fraction of H_2	0.013
Mass fraction of O	0.012
Mass fraction of OH	0.092

Table 3: Inlet boundary conditions

The far-field and nozzle wall boundaries were both defined as wall-type conditions, while the axis_sym boundary was defined as an axis, to correctly represent the 2D axisymmetric geometry of the problem.

Finally, the operating conditions were adjusted by setting the operating pressure equal to the local atmospheric pressure at the rocket’s altitude. The first operating pressure was set to 90001 Pa in the first scenario, corresponding to 1 km of altitude and at 30977 Pa at 10 km in altitude in the second scenario in the case study 1.

3.3.6 Solution Setup and Convergence

The last step in setting up the CFD simulation is the configuration of the solution. At this stage, a numerical method to solve the governing equations must be chosen. I used the implicit ROE-FDS, which is well-suited for compressible, supersonic flows such as rocket exhaust plumes.

A standard initialisation was applied to compute from the inlet to provide a stable starting point for the solution.

Finally, the simulation was run for 4000 iterations until a satisfactory level of convergence was reached, ensuring that the residuals decreased consistently and the monitored physical quantities stabilised.

3.3.7 Visualisation Strategy

Now that the simulation setup is complete and the model is running, the next crucial step is visualisation. This aspect is often undervalued, yet it plays an important part in both understanding and communicating the results. Having numerical outputs is not enough. Visualisation allows us to interpret them, identify patterns, and validate whether the simulation makes sense. It also provides an effective way to communicate findings without long explanations.

For the first case study, I chose to visualise the contour of temperature, pressure, velocity and the distribution of each species. Parameters like pressure and velocity are quite important, especially as they serve to check if the results are physically consistent and realistic.

For the second case study, no graphical visualisations are provided because the simulations were already computationally demanding. Instead, the analysis focuses on numerical results and their interpretation to evaluate the case.

3.3.8 Parameter set

To gain insight into rocket emissions, I modelled the ascent through the troposphere. For this, I used the parameter set tool, where parameters and their values can be defined either as constants or expressions. In my case, I created two input parameters: `pressure_atm` is defined in the operating conditions, and `temperature_atm` is defined at the outlet. In addition, I generate one output parameter, the mass flow rate, evaluated at four altitude points. To calculate the input values, I used the equations introduced in the theoretical background section. The results are shown in Table 4.

Altitude (m)	Pressure (Pa)	Temperature (K)
1000	9001	281.5
3000	71009	268.5
6000	49763	249
10000	30977	223

Table 4: Values of parameter set

However, due to the high computational cost and long runtime, I limited the study to a single output parameter. This choice allowed me to gain a first understanding of the computational demands before attempting a full simulation.

3.4 Trials and errors

After setting up the simulation, I had to run several trials, as the first attempts did not give satisfactory results. Many hours and considerable effort were spent identifying the sources of error and correcting them. The initial objective was simply to obtain at least one converged solution (without parameter sweeps), which required trials and adjustments.

In the very first trial, the solver diverged, which was not surprising, as it is rare for a complex CFD case to converge on the first attempt. The results of this run are shown in Figures 10, 11, and 12, which present the scaled residuals, pressure distribution, and temperature field, respectively.

Almost all monitored quantities increased monotonically, except for $k - \omega$. Moreover, the pressure and temperature fields did not exhibit the expected patterns, and several peak values were physically unrealistic. These inconsistencies indicated that the setup did not represent the physical problem correctly.

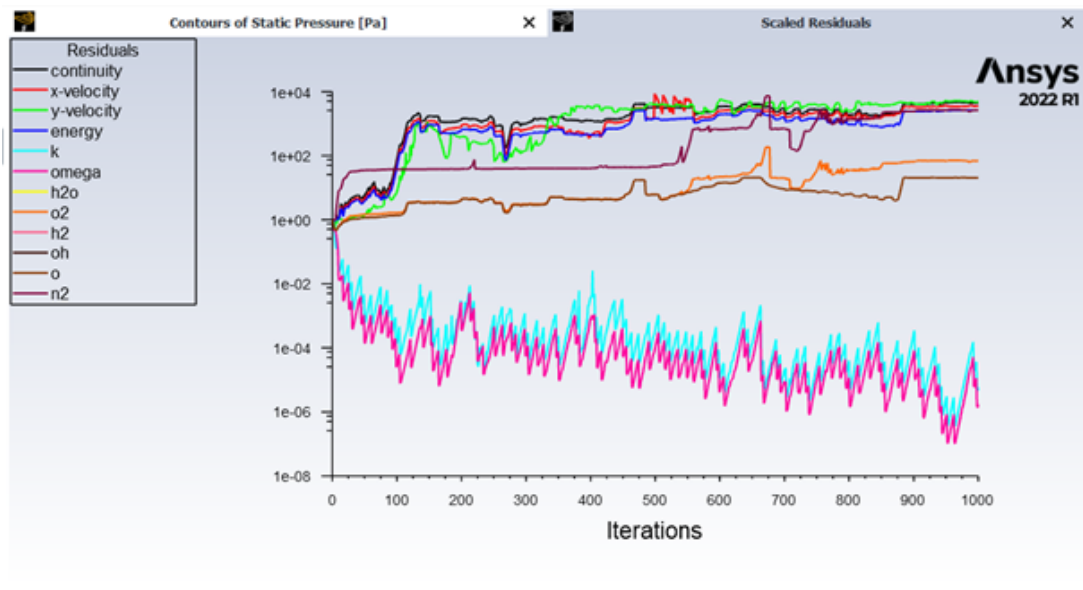


Figure 10: Scaled residuals from the first simulation

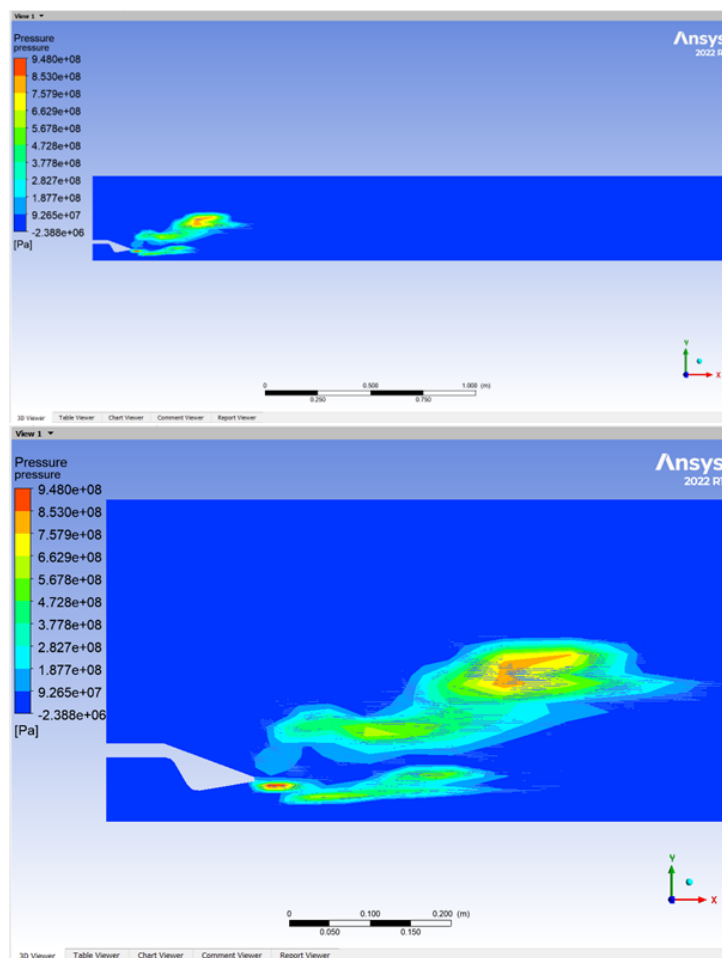


Figure 11: Pressure from the first simulation

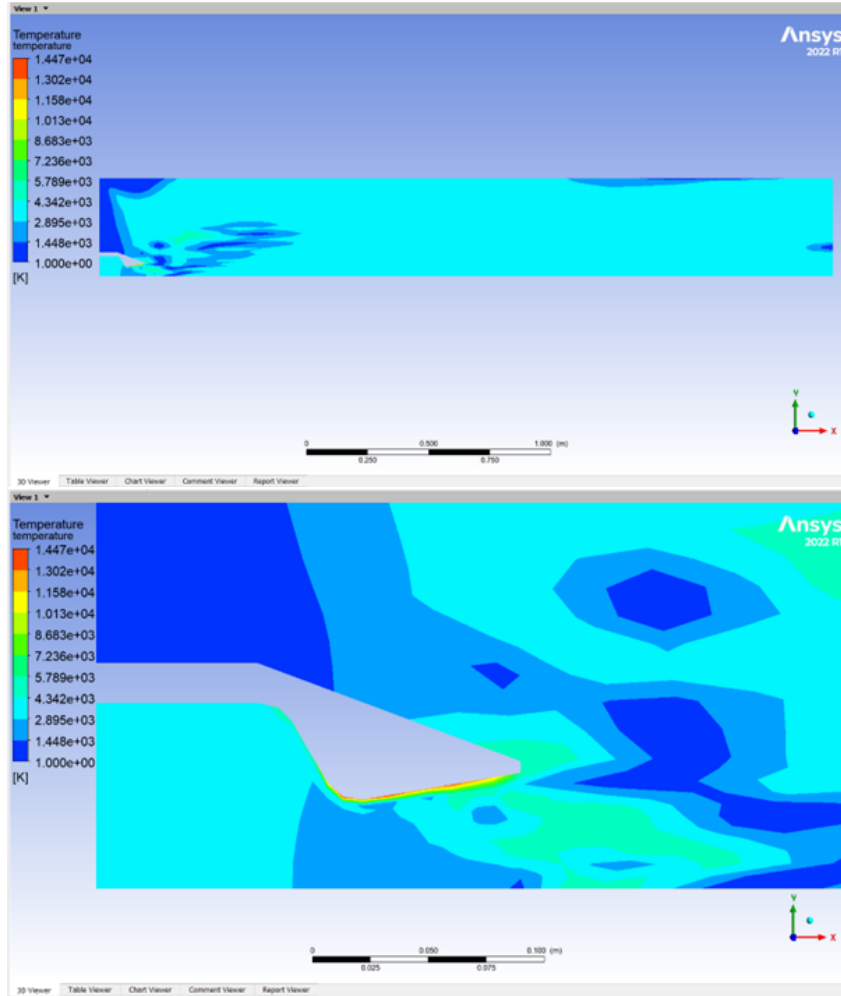
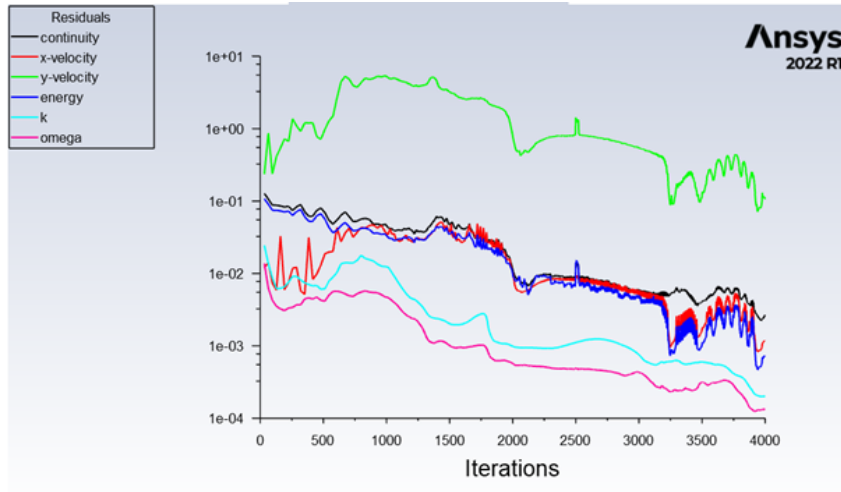


Figure 12: Temperature from the first simulation

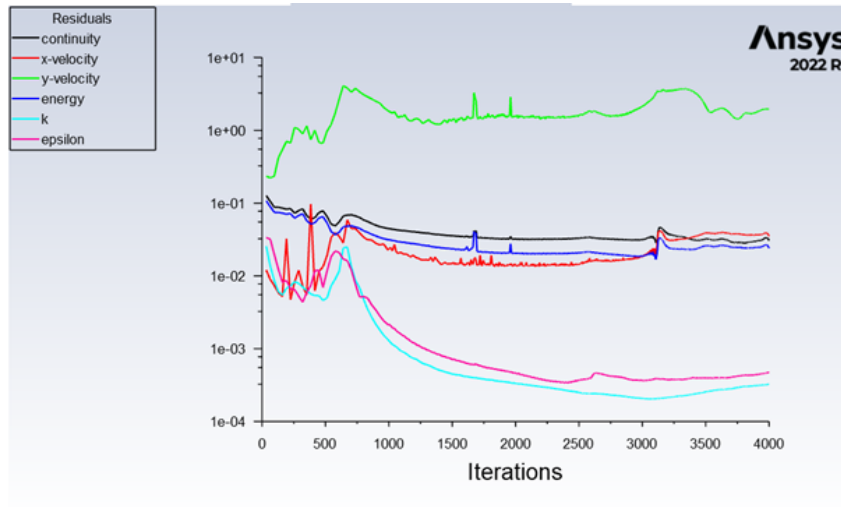
Upon review, the first major issue was the geometry. The initial geometry was not optimal, and at one point, I also made a unit conversion error. The second issue concerned the mesh. The first mesh was too coarse to resolve key gradients, and its quality metrics (aspect ratio and orthogonal quality) were close to their lower acceptable limits. As a result, the solution lacked both accuracy and stability. After correcting the geometry scale, redefining the domain to start at the throat, and refining the mesh, the solution behaviour became much more consistent with physical expectations.

Another important source of error was the boundary conditions. These were challenging to define correctly, and several mistakes occurred with the pressure, temperature, and boundary types. For example, at one stage, I mistakenly defined the far-field as a pressure outlet instead of a wall. This caused immediate divergence, with the continuity residual rising sharply. In reality, the far-field should not be represented as a solid wall either; however, due to time constraints, I could not fully resolve the divergence issue that appeared when modelling it as a pressure outlet. A working hypothesis is that the computational domain was too small, though this requires further testing.

During the setup, I also tested different turbulence models. In particular, I compared the $k - \epsilon$ and $k - \omega$ models (Figures 13 and 14).



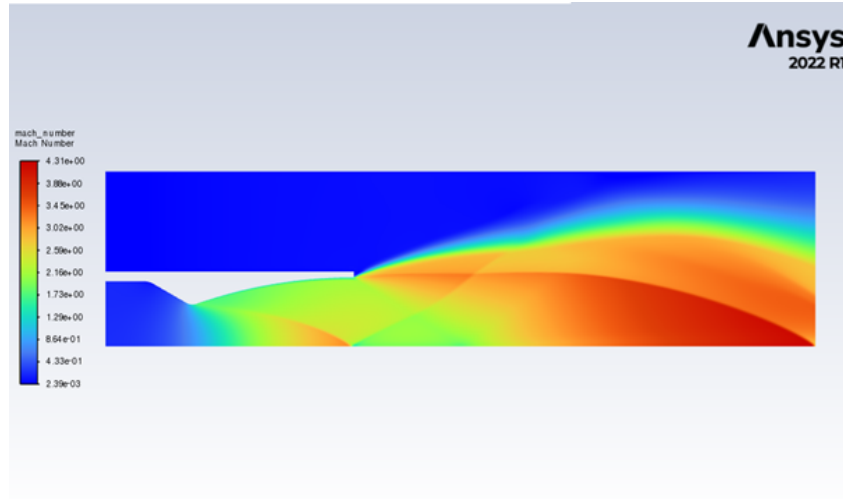
(a) Scaled Residuals with model k - ω



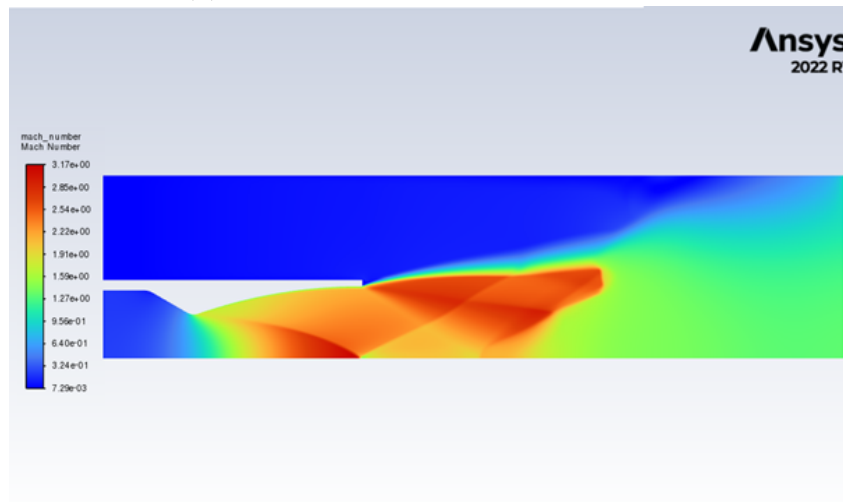
(b) Scaled Residuals with model k - ϵ

Figure 13: Difference in the scaled residuals between two models

The scaled residuals showed that the $k - \epsilon$ model produced a more stable trend, although convergence was much slower, requiring significantly more iterations.



(a) Mach number with model k-omega



(b) Mach number with model k-epsilon

Figure 14: Difference in the Mach number between two models

The Mach number field highlighted the differences more clearly: with the $k - \omega$ model, the exhaust behaviour aligned better with theoretical expectations, while the $k - \epsilon$ model produced inconsistencies, particularly near the end of the plume where the Mach number varied abruptly instead of smoothly.

In summary, the process involved extensive trial and error before reaching a case that produced coherent and physically reasonable results.

Step by Step Approach

Another recurring mistake I made was trying to move too quickly. I was eager to obtain results as soon as possible, but this approach actually cost me more time in the long run. For example, I initially attempted to model the entire system directly, including all species in the boundary conditions. This quickly became overwhelming due to the multiple parameters and the inability to find the errors. It led to significant delays before I realised the complexity of the setup.

The best approach was to proceed step by step. I learned to begin with a simplified model, for instance, by disabling species transport and using air as the working fluid. Once this baseline simulation worked, I could gradually introduce more complexity, changing one parameter at a time. This incremental method made it easier to identify what was blocking the simulation and how to resolve it.

Since this was a type of problem I had not encountered in previous projects, I also had to account for new parameters and interactions that could interfere with convergence. This forced me to adapt my approach.

4 Results and Discussions

In the context of the exponential growth of the space industry, this study aims to investigate how rocket emissions interact with the atmosphere during ascent through the troposphere. To achieve this, the nozzle exhaust plume was modelled after combustion to analyse the distribution of species in the exhaust. Firstly, two static cases were considered: one at an altitude of 1 km and the other at 10 km of altitude, to provide a baseline comparison. Expanding on these initial cases, a parameter set was used to simulate the continuous ascent of the rocket through the troposphere.

4.1 Case Study 1: Rocket Exhaust at 1 km altitude

The first case focuses on modelling the rocket exhaust at 1 km from the surface to establish an initial understanding of the system. In this section, key performance parameters such as pressure and temperature are examined to assess their values and verify whether they remain physically realistic. Following this, the distribution and interactions of the different species within the plume are analysed to gain deeper insight into the underlying combustion products and their behaviour in the near-field region.

4.1.1 Convergence Behaviour of the Simulation

The first step is to evaluate how well the simulation performed. The run required approximately one and a half hours to reach the maximum number of iterations. Through the multiple iterations, the residuals showed a general trend toward convergence, although full convergence was not achieved. As shown in Figure 15, the process displayed oscillations throughout. After 4,000 iterations, the continuity residual settled around 10^{-3} . The main exception was the energy residual, which initially increased during the first 1,000 iterations before gradually decreasing. The sudden spike at 2,700 iterations is not the result of physical readjustment, where the trend continued to converge.

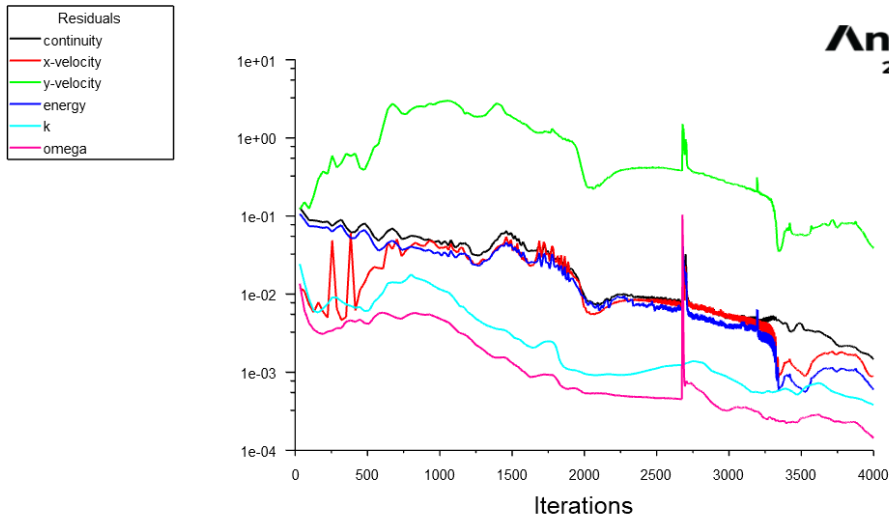


Figure 15: Case study 1: scaled residuals at 1 km

A positive point is that the simulation managed to converge, even if it was not fully stable and did not reach the ideal value of 10^{-4} . The unusual trend observed in the energy residual is most likely caused by the high temperatures combined with species transport. Although not perfect, this result can still serve as a useful starting point for further work.

4.1.2 Pressure Distribution in the Nozzle

The next step is to examine the pressure distribution. Figure 16 illustrates the results. As expected, the highest pressure is concentrated upstream of the throat, which is consistent with theoretical predictions: the gases must reach high pressure before accelerating through the throat to generate thrust. Downstream of the throat, the pressure drops significantly as the flow expands.

At the nozzle exit, a distinct diamond-shaped pattern can be observed. These are shock and expansion waves, which are characteristic features of supersonic nozzle flows. Their presence indicates that the model correctly reproduces the physical behaviour of a rocket exhaust plume. The shocks also suggest that the nozzle operates under slightly off-design conditions, which is normal when simulating fixed ambient pressure at a given altitude.

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Figure 16: Case study 1: Pressure distribution at 1 km

4.1.3 Temperature Distribution in the Nozzle

Regarding the temperature distribution, the results show a richer variation compared to the pressure field, with a broader range of gradients. The overall pattern is that very high temperatures are concentrated in the combustion chamber and throat, followed by a gradual decrease downstream through the nozzle. In the exhaust plume, there is a mixing of temperatures outlining the different temperatures wherever there is a shock or an expansion wave. These variations are important as they illustrate how the exhaust gases interact dynamically with the surrounding atmosphere.

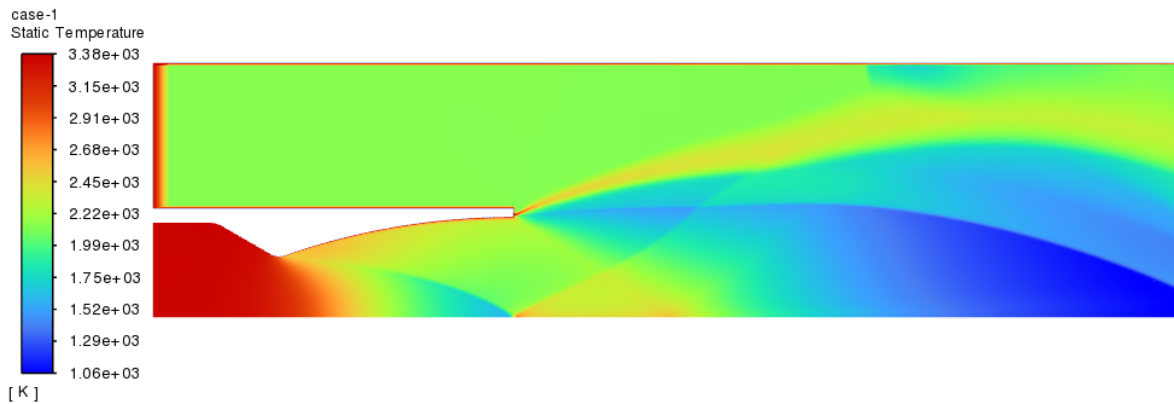


Figure 17: Case study 1: Temperature distribution at 1 km

Another notable feature is the thin high-temperature band along the nozzle wall and the fairfield border. This corresponds to heat transfer from the combustion gases to the wall, which in real rocket engines can contribute to material stress and erosion. Identifying this behaviour in the model is valuable, as it demonstrates that the simulation captures not only plume expansion but also wall-gas thermal interactions. However, this characteristic shouldn't be present near the fairfield. This is because I set the fairfield as a wall in the boundary conditions. In reality, it should be a pressure outlet, but the simulation diverged and the type changed.

4.1.4 Velocity Distribution in the Nozzle

To better analyse the velocity distribution within the geometry, the Mach number was examined, as it provides a clearer representation of the flow behaviour. As expected, the Mach number increases significantly after the throat, reflecting the transition from subsonic to supersonic flow. In the combustion chamber, the Mach number remains below 1, while in the nozzle it rises above 1, consistent with theoretical predictions.

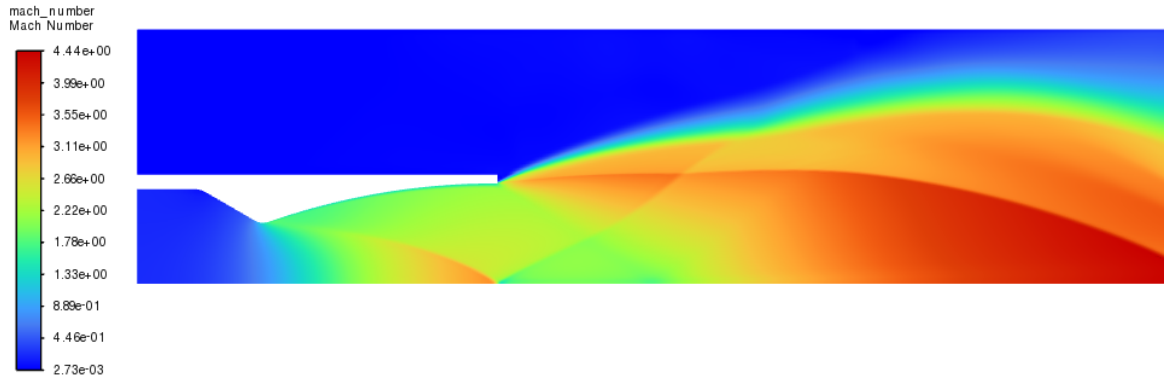


Figure 18: Case study 1: Mach number distribution at 1 km

The results also reveal the role of shock and expansion waves in the exhaust plume. Shock waves appear as regions of reduced Mach number, corresponding to local compression of the gases, while expansion waves are associated with regions of higher Mach number, where the gases accelerate and pressure and temperature decrease. This behaviour aligns well with the underlying physics of compressible flow and confirms that the simulation reproduces the expected dynamics of a rocket nozzle.

Overall, the analysis of pressure, temperature, and velocity provides a first validation of the simulation. The results reproduce mostly the expected physical behaviour of a rocket nozzle, particularly the formation of shock and expansion waves. While not yet fully converged, the trends confirm that the model captures the key dynamics of the exhaust plume and can serve as a reliable starting point for further refinement.

4.1.5 Species Distribution and Interaction

With the first analysis validated, the next step is to compare the different species and examine how they interact with each other in the exhaust plume.



(a) H_2O distribution



(b) N_2 distribution

Figure 19: Case study 1: H_2O and N_2 distributions at 1 km



(a) O_2 distribution



(b) O distribution

Figure 20: Case study 1: O_2 and O distributions at 1 km



(a) H_2 distribution



(b) OH distribution

Figure 21: Case study 1: H_2 and O_2 distributions at 1 km

One striking observation is that, among the six species considered, two distinct behaviours appear. The first category includes oxygen and nitrogen, which are more concentrated in the top right corner. The second category shows the opposite trend: water vapour, hydrogen, atomic oxygen, and hydroxyl are more concentrated in the nozzle and plume. These results differ from the expected behaviour, in which species composing the atmosphere are located in a small corner. This is principally the result of an error in the boundary conditions at fairfield. The species are primarily separated from each other. However, oxygen is set as a boundary condition for both atmospheric species and combustion gases. This is mainly due to the proportion of oxygen in the atmosphere being larger.

4.2 Case study 1: Rocket Exhaust at 10 km altitude

As the first scenario of case study 1, this second scenario models also a rocket exhaust, but at a higher altitude of 10 km. This second scenario serves to examine how the exhaust behaviour changes with altitude. The analysis will first consider performance parameters such as the Mach number to assess consistency with physical expectations. Afterwards, the distribution and interaction of the different chemical species defined in the model will be examined.

4.2.1 Convergence Behaviour of the Nozzle

The simulation was run for all 4,000 iterations, taking about one and a half hours to complete. The residuals illustrated in Figure 22, gradually decreased with small oscillations, reaching convergence around 1.10^{-3} . Similar to the 1 km of altitude case, the energy residual first increased before gradually decreasing at a pace comparable to the other monitored quantities.

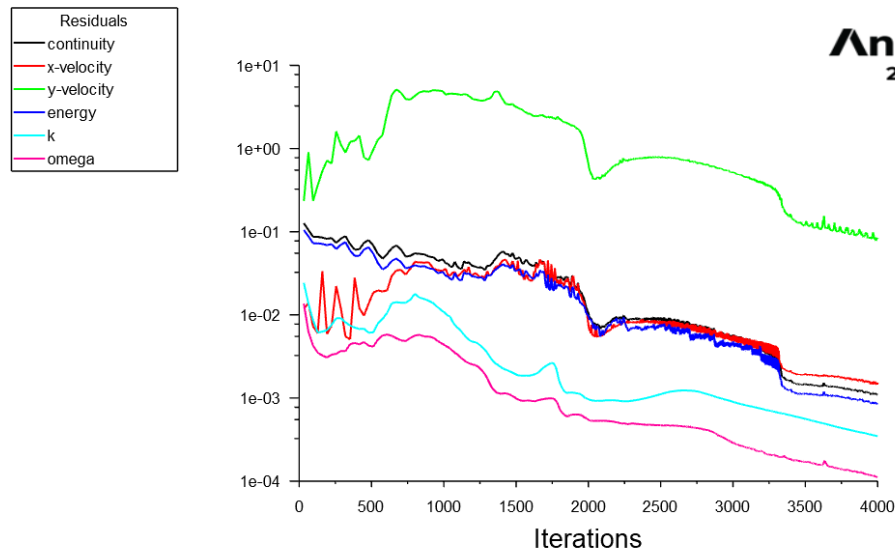


Figure 22: Case Study 1: Scaled residuals at 10 km

4.2.2 Pressure Distribution of the Nozzle

The pressure distribution (Figure 23) at an altitude of 10 km remains consistent with the expected physical behaviour inside the nozzle. The simulation still captures the presence of shock and expansion waves, confirming that the model represents the physics accurately. Compared to the first scenario conditions, the reduced ambient pressure slightly alters the wave pattern, as the nozzle operates closer to its optimal expansion. This effect, located at the exit of the nozzle, influences the position and intensity of the shock structures, which appear further downstream in the exhaust plume.



Figure 23: Case Study 1: Pressure distribution at 10 km

4.2.3 Temperature Distribution of the Nozzle

Concerning the temperature distribution (Figure 24), a wide range of values is still observed, clearly illustrated by the colour variations.

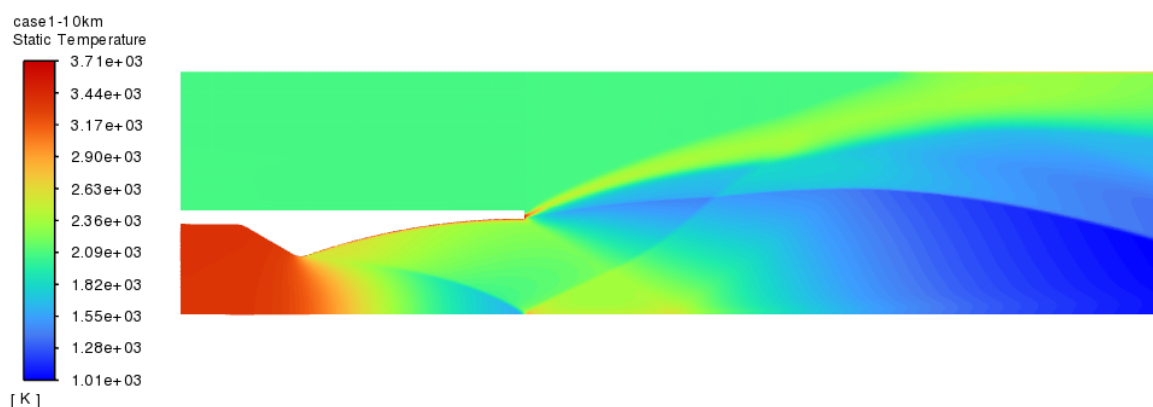


Figure 24: Case Study 1: Temperature distribution at 10 km

Compared to the first scenario, the shock and expansion waves appear more elongated, reflecting the influence of reduced ambient pressure at higher altitude. The most notable difference, however, is the absence of the thin layer of high temperature near the farfield boundary. This may be linked to the lower atmospheric density at 10 km, which reduces the intensity of local interactions, or it may indicate that such effects would only manifest on a longer timescale. As an aside, the behaviour should not have the thin layer due to an inaccurate boundary condition being applied to the far field. Therefore, due to the

lower density, the expected heat transfer behaviour of the wall may not be fully captured.

4.2.4 Velocity Distribution of the Nozzle

Regarding the velocity distribution at 10 km altitude, the Mach number provides a clear representation of the shock and expansion waves. As already observed in the temperature field, these waves appear more elongated downstream, reflecting the influence of lower ambient pressure. This behaviour is consistent with the expected physics and offers a clear visualisation of the plume dynamics.

Ansys
2022 R1

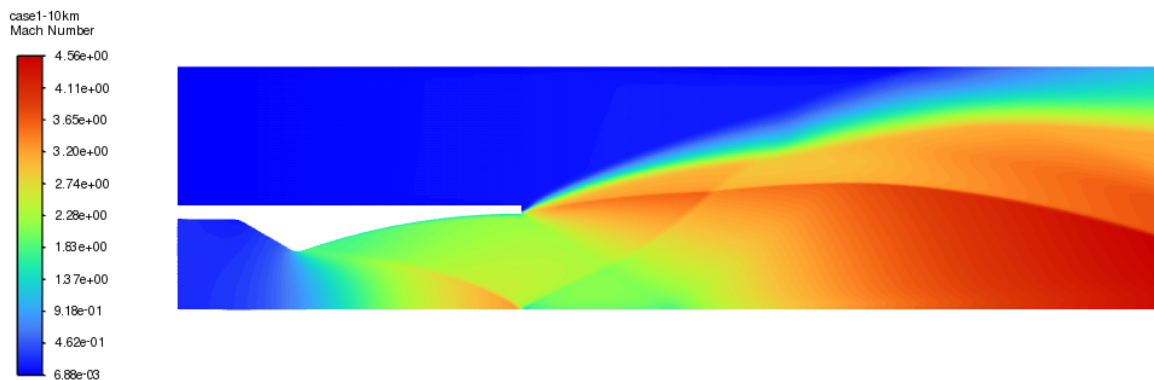


Figure 25: Case Study 1: Mach number distribution at 10 km

In summary, the results at 10 km altitude are consistent with the surface case while highlighting the influence of reduced atmospheric pressure. The nozzle flow maintains the expected physical behaviour, with visible shock and expansion waves that extend further downstream. Both the temperature and Mach number distributions confirm these effects. Overall, the simulation provides a coherent picture of exhaust plume dynamics at higher altitudes.

4.2.5 Species Distribution of the Nozzle

Regarding the species distribution, the results tell a different story. Unlike the pressure, temperature, and velocity fields, all species remain nearly constant throughout the domain. Figures 26, 27, and 28 show no visible pattern or interaction between the atmospheric components and the exhaust plume. This behaviour likely originates from limitations in the boundary conditions or the species transport setup, which may have prevented proper mixing between the exhaust gases and the surrounding atmosphere. Another possible factor is insufficient mesh resolution in the plume region, which can smooth out gradients and suppress the expected variations.

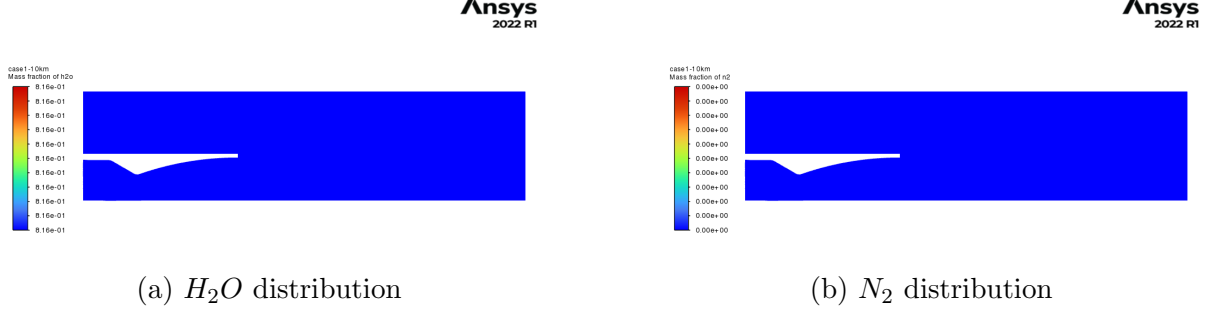


Figure 26: Case study 1: H_2O and N_2 distributions at 10 km

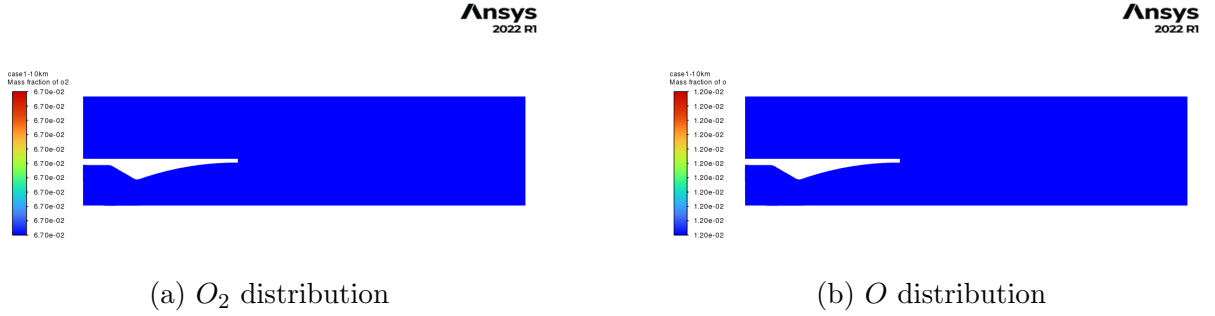


Figure 27: Case study 1: O_2 and O distributions at 10 km

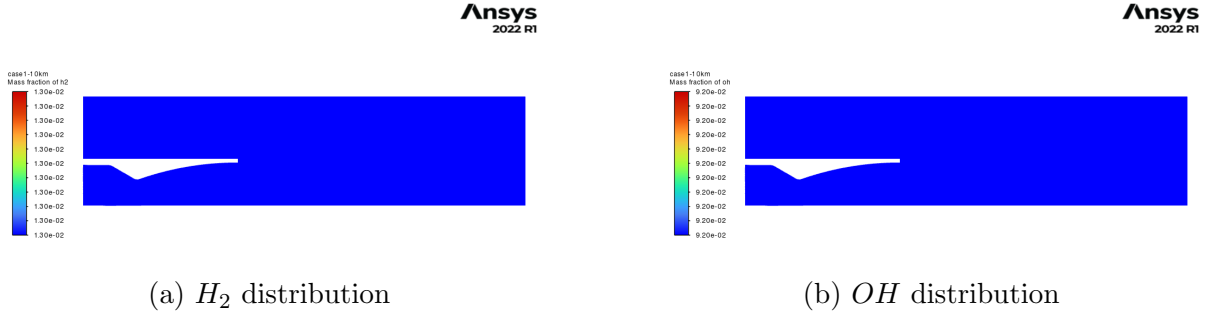


Figure 28: Case study 1: H_2 and O_2 distributions at km

4.3 Comparative Analysis of Case Study 1

Reviewing the two scenarios of Case Study 1, both similarities and differences can be observed.

First, the solver in both simulations required the maximum number of iterations to reach even a minimal level of convergence, which made the process time-consuming. This highlights the computational complexity of such simulations and underlines the importance of carefully defining the objective: one must strike a balance between simplifying the model and still obtaining realistic results.

Second, the pressure distribution was very similar in both cases. The highest values were concentrated in the combustion chamber, with limited interaction with the surrounding atmosphere. The main difference between the two altitudes lies in the values rather than the pattern: the maximum pressure decreased by roughly 40,000 Pa at 10 km compared to 1 km, which is consistent with the lower ambient pressure at higher altitudes.

Third, the temperature analysis revealed more contrast. At 1 km altitude, the results showed the effect of a boundary condition error that introduced an unrealistic thin layer of high temperature near the farfield. At 10 km, this layer disappeared. A possible explanation is that the lower density of the atmosphere at higher altitudes reduces the rate of interaction between the exhaust gases and the environment, meaning that such thin layers would take more time to develop and are therefore not captured under the current setup.

Fourth, the velocity field gave the most realistic results in both cases, with a clear supersonic flow and visible shock and expansion waves. At higher altitude, the exhaust plume appeared longer downstream, which aligns with expectations due to reduced ambient pressure allowing the jet to expand further.

Finally, the species distribution results were more intriguing. At 1 km, there was a clear contrast between atmospheric species (oxygen and nitrogen) and exhaust products (water vapour, hydrogen, hydroxyl, and atomic oxygen). However, at 10 km, all species remained constant throughout the domain, with no visible interaction between atmosphere and exhaust. A closer comparison of maximum values revealed differences only in the atmospheric species:

Altitude	O_2	N_2
1 km	$1.32e - 1$	$3.57e - 1$
10 km	$6.70e - 2$	$0.00e0$

Table 5: Table showing the differences between the two scenarios and the species involved

One hypothesis is that at higher altitudes, the concentration of nitrogen becomes negligible, while oxygen remains present partly because it is also a combustion product. This could explain why the exhaust and atmospheric compositions appear more homogeneous at 10 km, differing only in proportion rather than in kind.

Overall, case study 1 provided a first consistent perspective of rocket exhaust behaviour in the lower atmosphere. The analysis demonstrated that pressure and velocity distributions behave largely as expected, while temperature and species distributions highlight both numerical challenges and potential physical explanations. These results validate that the model can capture the main features of nozzle flow and plume expansion, but also point to areas where improvements in boundary conditions and domain definition are required.

Comparison with Existing Literature

The results from case study 1 can be related to several findings from previous studies on rocket nozzle flows and exhaust plume behaviour. In both scenarios, the formation of shock and expansion waves inside and outside the nozzle corresponds well with the classical theory of supersonic nozzles. It has been widely reported in both experimental and numerical studies [1]. The presence of diamond patterns of the shock and expansion waves that elongate with the altitude is constant. Previous research shows that lower ambient pressure causes the plume to expand further downstream [29].

The pressure and velocity distributions observed in this study are validating CFD analyses and experimental measurements conducted in wind tunnels and flight conditions. Typically, reports high pressures before the throat and supersonic acceleration in the

divergent section [18]. Likewise, temperature distributions show high values in the combustion chamber and a progressive decrease in the exhaust plume.

However, the gap appears when modelling chemical species. At 10 km altitude, the simulation results show constant species distributions across the geometry, with little evidence of mixing between exhaust gases and the ambient atmosphere. Previous work highlights that accurate prediction of exhaust-atmosphere interactions requires careful treatment of boundary conditions, larger computational domains, and advanced turbulence–chemistry interaction models [4]. These aspects are not fully captured in the present setup, which likely explains the simplified results observed at higher altitudes.

Overall, while the results obtained reproduce key aerodynamic features, further refinements are required to improve the accuracy of species interactions and validate the model against experimental datasets.

4.4 Case study 2: Rocket Exhaust Ascension through Troposphere

For case study 2, a more complex simulation was developed by recreating the same model at different altitudes in the troposphere, adjusting the atmospheric pressure and temperature accordingly. Since a single simulation was already computationally heavy, only one output parameter was selected. The purpose of this case study was to evaluate how demanding the parameter set tool is in terms of computation.

Each scenario, covering four altitude points, required approximately one and a half hours to complete—similar to the runtime of a single standalone simulation.

The results are shown in Table 6.

Altitude	pressure_atm	temperature_atm	mass_flow_rate-op
1 km	90001 Pa	281.5 K	6.5636 $kg.s^{-1}$
3 km	71009 Pa	268.5 K	6.5636 $kg.s^{-1}$
6 km	49764 Pa	249 K	6.5636 $kg.s^{-1}$
10 km	39077 Pa	223 K	6.5636 $kg.s^{-1}$

Table 6: Results of complete simulation of the ascension of the rocket

The mass flow rate remained constant across all simulations. This result is expected, since the inlet conditions were unchanged with altitude. A constant flow, therefore, reflects a consistent simulation setup rather than a physical variation.

Returning to the computational cost, the results clearly highlight that increasing the number of output parameters would significantly extend the runtime. A potential solution to reduce this cost could be to optimise the mesh resolution, simplify the geometry, or test reduced-order modelling approaches before running full-scale simulations. A study by Vestnes (2016) also involved modelling the exhaust plume of a rocket, addressing the difficulty of achieving converging results due to the complexity of the problem ??.

In summary, while Case Study 2 confirmed the heavy computational cost of using parameter sets, it also underlined the importance of carefully balancing accuracy and efficiency. Unlike case study 1, which provided detailed insight into flow parameters and

species interactions, this second case study highlighted more of the practical limitations of numerical modelling. Together, the two cases provide a complementary view: one focusing on physical understanding, and the other on computational feasibility.

5 Conclusion

With the rise of a new era in the space industry and the limited knowledge concerning the environmental impact of rocket launches, this study aimed to provide an initial insight into the modelling of rocket emissions during liftoff in the troposphere. Two case studies were conducted in Ansys Fluent.

The first case focused on modelling the exhaust plume of a rocket using liquid hydrogen and liquid oxygen as propellants, to observe how the exhaust plume interacts with the atmosphere. The results showed acceptable behaviour with the expected physical phenomena, including the presence of shock and expansion waves. However, two important issues were identified. The first was related to the boundary condition at the farfield: while it prevented divergence, it also introduced unrealistic effects at lower altitudes. Interestingly, this error became less visible at higher altitudes, where the lower atmospheric density reduces plume–air interactions. The second issue concerned the species distribution. While atmospheric and exhaust species could be distinguished at low altitude, at higher altitude the distribution appeared uniform, raising questions about how the model handles interactions under reduced pressure conditions.

The second case extended the first by simulating the rocket plume across several altitudes in the troposphere using parameter sets. However, the heavy computational cost quickly became evident. Even with only one output parameter (mass flow rate), each scenario required significant computation time, making it impractical to expand the study to multiple outputs. This highlighted a trade-off between model accuracy and computational feasibility.

Overall, this work demonstrated that while CFD can capture the key physical mechanisms of rocket exhausts, careful attention must be paid to boundary conditions, species interactions, and computational efficiency. These initial findings provide a foundation for future studies aiming to refine the models and, ultimately, to improve our understanding of the environmental impact of rocket emissions.

6 Future Work

In continuing this research, several adjustments could be made to enhance the simulations, deepen the understanding of rocket emissions and correct mistakes.

The first avenue concerns improving the simulation framework itself. One option would be to implement User Defined Functions (UDFs) in Ansys, which allow custom programming in C. Instead of relying on fixed parameter sets, UDFs would enable continuous variation of parameters during the simulation, leading to more realistic and adaptable results. This could be complemented by the use of a dynamic mesh, which better captures the evolving flow field during the rocket’s ascent. Furthermore, to obtain realistic trajectory coordinates, the open-source Python library RocketPy could be integrated. RocketPy is designed to handle both simple cases—with minimal input parameters—and more complex scenarios involving detailed specifications of the rocket, motor, and atmospheric environment.

The second possible extension would involve modelling the combustion chamber before ignition, i.e., introducing the liquid fuel and oxidiser as inputs and simulating their mixing

and reaction directly. This would require a multiphase, pressure-based approach. A similar attempt was made by Vestnes in 2016, but the study reported difficulties in achieving convergence. With advances in Ansys and CFD techniques since then, it would be valuable to revisit this scenario to assess whether improved solvers and computational resources could yield successful results.

A third line of investigation would be to evaluate the influence of different propellants. By running the same exhaust plume simulations with alternative fuel–oxidiser combinations, it would be possible to compare their relative impacts. The selection of propellants should focus on those that are most commonly used today and those expected to dominate in the near future, providing insight into the environmental trade-offs of evolving launch technologies.

Finally, the scope of the study could be expanded to link the exhaust plume simulation with atmospheric dispersion models. This would allow an assessment of how the local emissions from a launch propagate and affect the broader environment. One of the most widely used tools for this purpose is GEOS-Chem, an open-source atmospheric chemistry model. While current literature suggests that rockets represent a relatively small share of global emissions, the projected increase in launch frequency could lead to significant cumulative effects if the issue is not properly understood and addressed through regulation.

7 Acknowledgements

First, I would like to thank Velio Space Consulting and Engineering, the company where I am doing my internship and where the foundation of my work lies.

Second, I would also like to thank Professor Jenkins, my thesis supervisor. Thank you for your advice and for helping me through all the steps involved in creating this thesis.

Finally, I would like to thank Cranfield University for giving me the opportunity to work on such a large project and for providing the necessary resources to complete this work.

The thesis was built using software that served as the technical basis for this report. I wrote the report using Overleaf, and I used Grammarly and ChatGPT to help with the proofreading and editing, ensuring the text was clear and fluent.

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8 Appendix

8.1 Example of the output of a CEA simulation

```
*****

      NASA-GLENN CHEMICAL EQUILIBRIUM PROGRAM CEA2, FEBRUARY 5, 2004
      BY  BONNIE MCBRIDE AND SANFORD GORDON
      REFS: NASA RP-1311, PART I, 1994 AND NASA RP-1311, PART II, 1996

*****

### CEA analysis performed on Sat 16-Aug-2025 17:10:42

# Problem Type: "Rocket" (Infinite Area Combustor)

prob case = _____7395 ro equilibrium

# Pressure (1 value):
p,bar= 50

# Oxidizer/Fuel Wt. ratio (1 value):
o/f = 8

# You selected the following fuels and oxidizers:
reac
fuel H2(L)           wt%=100.0000
oxid O2(L)           wt%=100.0000

# You selected these options for output:
# short version of output
output short
# Proportions of any products will be expressed as Mass Fractions.
output massf
# Heat will be expressed as siunits
output siunits

# Input prepared by this script:/var/www/sites/cearun/cgi-bin/CEARUN/prepareInputFile.cgi

### IMPORTANT:  The following line is the end of your CEA input file!
end
```

THEORETICAL ROCKET PERFORMANCE ASSUMING EQUILIBRIUM

COMPOSITION DURING EXPANSION FROM INFINITE AREA COMBUSTOR

Pin = 725.2 PSIA

CASE = -----

	REACTANT	WT FRACTION (SEE NOTE)	ENERGY KJ/KG-MOL	TEMP K
FUEL	H2(L)	1.0000000	-9012.000	20.270
OXIDANT	O2(L)	1.0000000	-12979.000	90.170

O/F= 8.00000 %FUEL= 11.111111 R,EQ.RATIO= 0.992085 PHI,EQ.RATIO= 0.992085

	CHAMBER	THROAT
Pinf/P	1.0000	1.7257
P, BAR	50.000	28.974
T, K	3546.04	3379.62
RHO, KG/CU M	2.7225 0	1.6776 0
H, KJ/KG	-857.26	-1829.06
U, KJ/KG	-2693.81	-3556.21
G, KJ/KG	-56936.7	-55276.6
S, KJ/(KG)(K)	15.8147	15.8147
M, (1/n)	16.054	16.269
(dLV/dLP)t	-1.04454	-1.03971
(dLV/dLT)p	1.7798	1.7302
Cp, KJ/(KG)(K)	10.3614	10.1271
GAMMAS	1.1284	1.1253
SON VEL,M/SEC	1439.6	1394.1
MACH NUMBER	0.000	1.000

PERFORMANCE PARAMETERS

Ae/At	1.0000
CSTAR, M/SEC	2137.9
CF	0.6521
Ivac, M/SEC	2633.0
Isp, M/SEC	1394.1

MASS FRACTIONS

*H	0.00195	0.00165
H02	0.00038	0.00026
*H2	0.01414	0.01282

H2O	0.79360	0.81514
H2O2	0.00007	0.00004
*O	0.01467	0.01209
*OH	0.10452	0.09142
*O2	0.07066	0.06658

* THERMODYNAMIC PROPERTIES FITTED TO 20000.K

NOTE. WEIGHT FRACTION OF FUEL IN TOTAL FUELS AND OF OXIDANT IN TOTAL OXIDANTS

8.2 Ethical CURES



7 August 2025

Dear Miss Pourrain ,

Reference: CURES/26243/2025

Project ID: 29395

Title: Study and numerical modelling of the atmospheric impact of experimental rocket launches.

Thank you for your application to the Cranfield University Research Ethics System (CURES).

We are pleased to inform you your CURES application, reference CURES/26243/2025 has been reviewed. You may now proceed with the research activities you have sought approval for.

If you have any queries, please contact CURES Support.

We wish you every success with your project.

Regards,

CURES Team